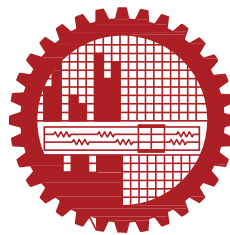


**CHOICE MODELING OF DWELLING ARRANGEMENTS
AND MOBILITY UNDER AFFORDABILITY AND
ACCESSIBILITY CONSTRAINTS IN DHAKA CITY**

ISMAMUL HOQUE

MASTER OF SCIENCE IN CIVIL ENGINEERING (TRANSPORTATION)



DEPARTMENT OF CIVIL ENGINEERING
BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

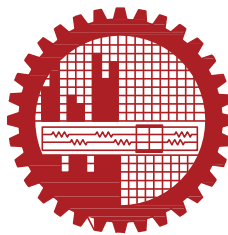
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The thesis titled “**Choice Modeling Of Dwelling Arrangements And Mobility Under Affordability And Accessibility Constraints In Dhaka City**”, submitted by **Ismamul Hoque**, Student ID: 0424042426, Session April 2024, has been accepted as satisfactory in partial fulfillment of the requirements for the degree of Master of Science in Civil Engineering (Transportation) on 15 June 2026.

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This is to certify that the work presented in this thesis titled, “Choice Modeling Of Dwelling Arrangements And Mobility Under Affordability And Accessibility Constraints In Dhaka City”, is the outcome of the research carried out by **Ismamul Hoque** under the supervision of **Dr. Sk Md Mashrur**, Assistant Professor, Department of Civil Engineering, Bangladesh University of Engineering and Technology (BUET), Dhaka-1205, Bangladesh.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma, or other qualification.

Ismamul Hoque
Student ID: F0424042426

DEDICATION

To my beloved family and my supervisor

ACKNOWLEDGEMENT

First and foremost, I express my deepest gratitude to the Almighty Allah, the most merciful and the most beneficent, for granting me the strength, patience, and perseverance necessary to complete this thesis work successfully.

This research required the application of engineering judgment, analytical thinking, and references from textbooks and previous related publications. The entire work has been conducted under the Department of Civil Engineering.

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ABSTRACT

Rapid urbanization in growing megacities is placing increasing pressure on already-strained housing and transportation systems. To support coordinated urban planning, it is essential to understand the factors influencing individuals' interconnected decisions, including where they live, how they travel, and where they conduct their daily activities. However, most existing research on emerging cities treats residential, household, and travel choices separately, thereby limiting integrated planning. Hence, this study endeavors to address this gap by jointly analyzing these interdependent decisions within a rapidly growing urban context.

Using data from 927 households across Dhaka, Bangladesh, a joint discrete-continuous model is developed. The model simultaneously estimated a 12-alternative mode-choice model, a 4-alternative living arrangement model, and a continuous model of commute distance to the Primary Activity Location (PAL). The latter two models accounted for modal accessibility and were linked through a copula-based correlation structure.

Results show that living arrangements are governed by life cycle and family obligations. Unmarried adults overwhelmingly remain within family households; the presence of multiple seniors strongly drives joint family living, and multiple earners enable independent nuclear households. Commute distance is lengthened most by motorization and the pursuit of affordable housing. License holders commute 1.2–1.7 km farther, and families with children in the dominant mid-rent band live about one kilometer farther from their PALs, while marriage and multiple earners pull households closer. Modal accessibility significantly reduces commute distance for households in high-density and residential neighborhoods and, most uniformly, for those willing to use the MRT within walking distance, although joint families' transit orientation collapses once stations lie beyond walkable reach. A positive correlation between the unobserved drivers of family-oriented living and longer commutes confirms the interdependence of the two choices.

Overall, the study provides actionable guidance for promoting affordable housing near PALs, walkable transit-oriented development, and housing supply responsive to diverse urban household needs.

LIST OF PUBLICATIONS

1. Hoque I, Naz A, Mashrur SM, Habib KN. Urban Choices in Megacities Unpacked: The Interplay of Housing, Accessibility, and Mobility Decisions. Accepted for presentation at the Transportation Research Board 2026 Annual Meeting; 2026 Jan 9–14; Washington, DC.
2. Hoque, I., Naz, A., Mashrur, S. M., & Habib, K. N. (2025). *Urban Choices in Megacities Unpacked: The Interplay of Housing, Accessibility, and Mobility Decisions*. Under review in a Q1 Journal.

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CHAPTER 1

INTRODUCTION

This chapter introduces the research presented in this thesis. It begins with the background and present state of the problem in Section 1.1, followed by the motivation of the study in Section 1.2. Section 1.3 states the goal and objectives, Section 1.4 defines the scope, and Section 1.5 outlines the organization of the thesis.

1.1 Background

Cities worldwide are facing the intertwined challenges of housing affordability and access to primary activity locations (i.e., employment and education), which are typically addressed in isolation leading to fragmented solutions (Guan & Peiser, 2018). Evidence indicates that in dual-income households, a marginal increase in collective accessibility (i.e., reflecting the perceived efficiency and practicality of available travel modes) is correlated with greater work–home dispersion and shorter aggregate commute distances (Hasnine et al., 2023). In addition, housing and transportation expenses frequently constitute 40% to 50% of household expenditures for low to middle-income countries (LMICs) (Dewita et al., 2020; Makarewicz et al., 2020). Hence, discerning the trade-offs among time, cost, and modal accessibility is critical but essential for developing equitable, integrated, and connectivity-driven urban growth strategies for LMICs.

With most urban households being renters in Dhaka (Haque et al., 2020), rising housing costs (i.e., a 5.9% rent increase from October to December of FY 2023–24) have placed a significant toll on urban residents (Giti, 2018). The severity is higher, particularly among low and middle-income households, which constitute 80% of Dhaka residents (“Bangladesh - Dhaka Improv. Living Cond. Urban Poor,” 2007). With private car ownership extremely low (i.e., only a few vehicles per 1,000 people), households rely heavily on walking, rickshaws, and buses, making transit and non-motorized accessibility crucial in shaping where families choose to live (Masoumi, 2021). Meanwhile, Dhaka’s chronic congestion exacerbates these challenges, with daily working hours lost on its roads increasing from 5 million in 2017 to over 8 million in 2022, costing the economy nearly Tk 140 crore per day (Jahan & Kalam, 2012). A handful of studies, with limited samples, explored how these mobility strains influence individual’s residential and mobility decisions in Dhaka, linking accessibility, affordability and daily travel behavior

(Choudhury & Ayaz, 2015b; Kamruzzaman & Ogura, 2008). Hence, this study endeavors to unpack such intricate interdependencies amongst multidimensional urban choices in emerging cities with rapidly growing economy.

These interdependencies surface in the everyday trade-off that Dhaka residents face between accessibility and affordability. Locations close to workplaces, educational institutions, and public transport generally offer better access to opportunities but carry higher rents, greater density, and limited dwelling space, while more affordable housing tends to lie farther from major activity centers, imposing longer commutes and heavier time and cost burdens (Dewita et al., 2020; Makarewicz et al., 2020). Who people live with adds a further layer to this trade-off. Unlike many Western contexts, South Asian cities exhibit a wide range of household configurations: individuals may live alone or with non-family members in shared accommodation, in nuclear families with or without children, or in multigenerational joint families, shaped by cultural norms, marriage, employment, and economic necessity (Hasnine et al., 2023; Mulder & Clark, 2002). Each arrangement carries different residential needs: a single worker may share housing with non-family members to remain in an accessible area at manageable cost, families with children weigh school access and dwelling space, and joint families may accept remote locations where larger homes are affordable (Pinjari et al., 2011).

The quality of available travel options matters as much as physical distance in these decisions. Two residents living equally far from their workplaces may experience entirely different daily realities depending on the modes available to them, the congestion they face, and the cost and reliability of their journeys (Akbari & Habib, 2015; Hasnine et al., 2019). Yet residential location, living arrangement, and daily travel have most often been studied as separate decisions, both internationally and in the limited Dhaka-specific literature (Schwanen & Mokhtarian, 2005; Zhang et al., 2019). This study addresses that problem: it examines how urban residents in Dhaka jointly decide whom to live with and how far from their workplace or educational institution, their Primary Activity Location (PAL), to reside, and how affordability pressures and the accessibility of available travel modes shape these intertwined decisions.

1.2 Motivation

The motivation for this research is, first of all, practical. Dhaka's housing and transportation systems are under mounting and simultaneous pressure: rapid urbanization has intensified the demand for residential space, while congestion and unequal transport access have made daily commuting increasingly burdensome (Dewita et al., 2020; van Heerden et al., 2022). For most residents, the decision of where to live is therefore also a mobility decision. Affordable housing located far from employment centers may reduce rent but inflate travel time and cost, while housing near workplaces may ease commuting yet remain out of reach for many households, a tension commonly discussed as location affordability, in which housing and transportation costs jointly determine how affordable a residential location truly is (Choudhury & Ayaz, 2015; Makarewicz et al., 2020). When housing and transport planning are poorly coordinated, these trade-offs deepen inequality in access to opportunities, especially for the city's low- and middle-income majority.

A second motivation is the diversity of household structures in Dhaka. Urban policies often treat households as uniform decision-making units, yet a single person sharing accommodation, a nuclear family with children, and a joint family spanning generations face very different residential needs and mobility constraints. The presence of children, the number of earners and students, income, vehicle access, and tenure status all influence how a household balances housing cost against access to daily activities (Haque et al., 2020; Hasnine et al., 2023). Policies that ignore this diversity, for instance, by supplying affordable housing in locations whose travel burdens few target households can bear, risk misjudging their own consequences.

Finally, the study is motivated by a gap in evidence. Because residential location, living arrangement, and travel behavior have largely been examined in isolation, planners in Dhaka currently lack integrated, behaviorally grounded evidence on how these decisions shape one another (Pinjari et al., 2011; Zhang et al., 2019). Producing such evidence for a rapidly growing megacity of the Global South, where affordability constraints, informal transport, and strong family obligations interact in ways rarely captured by studies of Western cities, is the central motivation of this thesis.

1.3 Goal and Objectives

The primary goal of the study is to provide clear, policy-ready evidence regarding the interplay of affordability, accessibility, and household structure while exploring the interlinked residential and mobility choices of Dhaka residents.

Specifically, the study objectives are:

- (i)** To assess variations in urban residents' home-to-activity distances across household types under affordability and accessibility constraints.
- (ii)** To evaluate the effect of household structure on modal choice, and access disparities in Dhaka's renter-dominant, multi-storey context.
- (iii)** To provide data-driven policy guidance considering the interplay between housing affordability, accessibility, and urban mobility in Dhaka.

1.4 Scope of the Study

This study is geographically confined to Dhaka city, Bangladesh. The analysis focuses on individuals whose residential location and Primary Activity Location are located within the relevant urban context of the study. The Primary Activity Location is defined as the main workplace or educational institution where the respondent spends a major portion of regular activity time.

The survey dataset includes socio-demographic characteristics, household attributes, housing information, vehicle availability, travel mode information, and geocoded residential and activity locations. The final analytical sample consists of 927 valid observations after excluding incomplete or inconsistent responses related to residential, workplace, and educational locations.

The research considers four dwelling arrangement categories: living alone or with non-family members, living with family without minors, living with family with minors, and living in a joint family arrangement. These alternatives represent the major household living patterns relevant to the urban and cultural context of Dhaka. The mobility component considers multiple travel modes, including public transport, private modes, paratransit, non-motorized modes and walking. The use of multiple modes allows the study to estimate a modal accessibility measure that reflects the attractiveness of available transport alternatives.

The methodological scope of the study includes a mode choice model, a living arrangement choice model and a lognormal distance model. The mode choice component is used to estimate modal accessibility through the expected maximum utility or logsum. This accessibility measure is then incorporated into the living arrangement and distance components. The joint structure captures the relationship between discrete dwelling arrangement choice and continuous residential distance from PAL.

The study primarily focuses on behavioral interpretation and policy relevance. It examines how accessibility, affordability, household composition, land use and mobility resources affect residential decisions. However, the study does not model the full housing market supply mechanism, real estate price formation or dynamic residential relocation over time. Since the analysis is based on cross-sectional survey data, the findings represent observed behavioral patterns at the time of data collection rather than long-term changes in household decisions.

1.5 Organization of the Thesis

This thesis is organized into five chapters, each addressing a specific component of the research.

Chapter 1	Introduces the background, motivation, objectives, and scope of the study.
Chapter 2	Literature Review on • Residential location choice, dwelling arrangement, and modal accessibility. • Housing affordability and commute distance in Dhaka. • Integrated land-use–transport and joint discrete–continuous modelling, and the research gap.
Chapter 3	Describes the study area, data, survey design, variables, and the joint modelling framework — mode choice, living arrangement, distance, and joint likelihood.
Chapter 4	Presents and discusses the estimation results, marginal effects, and the behavioral links among accessibility, affordability, household structure, and residential distance.
Chapter 5	Concludes with key findings, policy implications, limitations, and directions for future research.

Figure 1.1: Thesis structure

CHAPTER 1 presents the introduction of the study. It discusses the general context of urban residential and mobility decisions, background of the research, motivation, objectives, scope and organization of the thesis.

CHAPTER 2 reviews the relevant literature on residential location choice, dwelling arrangement, accessibility, affordability, commute distance, integrated land-use and transportation modeling and joint discrete-continuous modeling approaches. It also

identifies the major research gaps addressed in this study.

CHAPTER 3 describes the study area, data source, survey design, variables and methodological framework. It presents the structure of the mode choice model, living arrangement model, distance model and joint likelihood formulation used in the analysis.

CHAPTER 4 presents the model estimation results and discussion. It interprets the effects of socio-demographic characteristics, household structure, housing attributes, built environment variables and modal accessibility on living arrangement and distance choice. It also discusses the behavioral meaning of the estimated parameters and the relationship between affordability and accessibility.

CHAPTER 5 summarizes the major findings of the study and presents policy implications for housing, transportation and land-use planning. It also discusses limitations of the research and suggests directions for future studies.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview

This literature review examines residential location choice, dwelling arrangement, modal accessibility, and affordability as mutually dependent urban decisions rather than isolated outcomes. The central proposition emerging from the literature is that households do not first choose a dwelling and then independently decide how to travel. Rather, they weigh rent, tenure, dwelling size, household composition, access to work and education, neighborhood conditions, and the availability, cost, and quality of transport at the same time. This interdependence becomes especially important in rapidly urbanizing megacities, where a short physical distance does not necessarily imply low travel burden and where low housing cost may conceal high commuting time and transport expenditure (Akbari & Habib, 2015; Geurs & van Wee, 2004; Handy, 1997; Makarewicz et al., 2020).

The literature further shows that the Dhaka case is not simply an example of generic “jobs–housing mismatch.” It is a more complicated setting in which affordability constraints, heterogeneous mobility systems, strong family obligations, shared accommodation, and joint-family living patterns interact with intense congestion and uneven land-use development. Bangladesh-based studies have already established that housing affordability is a serious constraint for low- and middle-income households in Dhaka and that residential choice cannot be understood without considering access to employment and education (“Bangladesh - Dhaka Improv. Living Cond. Urban Poor,” 2007; *Google Earth*, n.d.; Choudhury & Ayaz, 2015b; Jahan & Kalam, 2012; Kamruzzaman & Ogura, 2008). However, this literature remains fragmented. Most studies analyze affordability, residential location, or travel separately; few explicitly model dwelling arrangement as a choice variable and very few use utility-based modal accessibility measures to explain how far households live from their PAL.

International research has advanced further by developing utility-consistent accessibility measures, joint discrete-choice frameworks, and integrated land-use–transport models. Yet the dominant evidence base remains heavily shaped by North American, European, or other relatively formalized urban settings, where household structures are often simplified and transport systems are more organized than in Dhaka (Akbari & Habib,

2015; Bhat & Guo, 2007; Hasnine et al., 2023; Pinjari et al., 2011). The present study therefore sits at the intersection of three underdeveloped strands in the literature: the modeling of living arrangement, the behavioral interpretation of modal accessibility and the affordability-driven variation in residential distance from workplace or educational anchors. Accordingly, this chapter starts with recent Bangladesh conditions, then reviews national and regional studies, then the wider global and theoretical literature, before identifying the research gap and setting out the conceptual framework for the study.

2.2 Bangladesh Context of Housing, Accessibility and Mobility

Dhaka provides an unusually strong case for studying the interdependence of housing and mobility choices. Census-based reporting indicates that Dhaka District had a population of 14.73 million in 2022, with roughly 76.26% of residents living in urban areas, while broader metropolitan reporting places Greater Dhaka at roughly 23.98 million people in 2024. These figures help explain why land, housing, and transport pressures are especially acute in the metropolitan region and why residential decisions are inseparable from accessibility concerns (BBS, 2022).

The transport environment of Dhaka is marked by chronic congestion, extreme modal heterogeneity, and severe competition for limited road space. Recent reporting describes average traffic speed in Dhaka at about 4.8 km/h, alongside a transport system in which buses, rickshaws, motorcycles, ride-hailing, cars, and pedestrians share the same corridors. At the same time, metro expansion has begun to alter the city's accessibility geography, with MRT Line 6 reportedly reaching around 400,000 daily passengers by February 2025. These conditions are directly relevant to residential behavior: they imply that the burden of a home–work or home–education separation depends not only on distance but also on mode availability, transfer burden, reliability, and first-mile/last-mile conditions (Shoukry, 2010).

Dhaka's land-use context amplifies this mobility problem. The city combines intense density, mixed land use, rapid spatial expansion, and fragmented development across formal and informal areas. Recent work on high-resolution land-use mapping for the Dhaka metropolitan area emphasizes that dense populations, diverse infrastructures, and rapid urban change make it difficult even to classify the city's land-use patterns consistently. For housing and mobility research, that complexity matters because accessibility is shaped not only by transport supply but also by where work, education,

retail, and services are located relative to households (Paul & Ali; *RAJUK Masterplan*).

The housing market in Dhaka remains under severe pressure. The literature consistently shows that households particularly low- and middle-income groups must navigate a difficult trade-off between rent or purchase cost on the one hand and commuting burden on the other. Formal housing near major employment and institutional centers is frequently unaffordable, while relatively cheaper housing is more available in peripheral or spatially less convenient areas (“Bangladesh - Dhaka Improv. Living Cond. Urban Poor,” 2007; *Google Earth*.; Jahan & Kalam, 2012; Kamruzzaman & Ogura, 2008). This means that affordability in Dhaka is best understood as location affordability rather than housing affordability alone. A dwelling that appears financially manageable in rent terms may become difficult to sustain once daily transport expenditure, time loss, congestion exposure, and household coordination costs are included.

This Bangladesh context is especially important for the present thesis because the city’s household arrangements are also more diverse than the structures commonly assumed in much of the international literature. In Dhaka, living separate alone or with non-family members, living with family without minors, living with family with children, and joint-family living all coexist within the same broad housing market. These arrangements imply different space needs, caregiving demands, school and work access needs, and cost-sharing possibilities. Thus, for Dhaka, the question is not simply where households live, but with whom they live and how those arrangements interact with mobility and affordability constraints.

2.3 Empirical Location Choice Evidence

2.3.1 Dhaka Affordability Location Evidence

Bangladesh based studies has established an important empirical base on urban housing affordability, particularly in Dhaka. Jahan & Kalam (2012) showed that rental affordability for middle-income households in Dhaka is under persistent strain, with many households required to allocate a disproportionately large share of income to rent. Kamruzzaman & Ogura (2008) likewise argued that the housing market serving middle-income households in Dhaka does not adequately match household needs in terms of price, space, and location. These studies are foundational because they demonstrate that affordability is not a marginal issue in Dhaka’s housing market, instead, it is a primary structuring constraint on residential options.

However, much of the early Bangladesh affordability literature relies on relatively static indicators such as rent-to-income ratios or purchase affordability thresholds. Such measures are useful, but they capture only one dimension of household burden. They do not fully represent the fact that a “cheap” dwelling may generate high transport spending or severe daily commuting fatigue. This limitation is important in Dhaka because housing and transport burdens are not spatially separable; they are jointly produced by the location of the dwelling relative to jobs, education, and available modes. In this sense, the contextual literature has accurately recognized affordability as a major issue, but it has only partially implemented the concept of affordability as a combined housing and mobility phenomenon.

Choudhury & Ayaz (2015) moved closer to a behavioral explanation by modeling residential location choice in Bangladesh and asking why households choose to live far from their key destinations. Their work is particularly relevant to the present study because it shifts the analysis from descriptive affordability toward observed location behavior. The implication of that work is that households may rationally accept larger distances when proximity is financially unattainable, when social and family obligations matter, or when other location attributes compensate for longer travel. However, the model does not account the influence of living arrangements and modal accessibility.

The Local evidence also suggests that residential choice in Dhaka cannot be interpreted without considering urban institutions and planning deficits. The World Bank’s Dhaka urban poor report emphasized the long-run mismatch between urban growth, infrastructure provision, and inclusive housing supply. That policy diagnosis remains relevant today: rapid growth without commensurate coordination among land use, transport, and housing institutions widens the gap between where opportunities are located and where households can afford to live (“Bangladesh - Dhaka Improv. Living Cond. Urban Poor,” 2007). This makes Dhaka a particularly relevant case for integrated behavioral analysis, because household decisions are being made in a context where the urban system itself is only partially coordinated.

2.3.2 Global South Housing Mobility Trade-Offs

Regional and comparable Global South studies reinforce the same broad lesson. Dewita et al. (2020) working on Indonesian cities, showed that transport affordability and housing affordability should be treated jointly, because households experience them as a combined budget burden. Their findings are highly relevant for Dhaka, where peripheral

housing may reduce rent but increase time and monetary travel cost.

A similar concern appears in South African scholarship. Venter et al. (2007) demonstrated that residential location and travel are deeply shaped by household inequality, gender relations, and transport disadvantage in low-income urban settlements. Later, van Heerden et al. (2022) argued that long-term spatial planning in developing-country contexts must jointly confront accessibility, affordability, and equity. The central implication for Dhaka is that spatial planning cannot rely on simple assumptions of market efficiency. If accessible locations remain unaffordable, the result is not just long commutes, but a socially unequal urban structure.

Evidence from Middle Eastern and other rapidly urbanizing metropolitan regions also points in the same direction. Masoumi (2021), comparing Istanbul, Tehran, and Cairo, found that commuting to work remains an important determinant of residential location, even in cities with highly complex urban forms. This matters for the Dhaka case because it suggests that the relevance of commute burden survives across very different cultural and institutional settings. At the same time, the form of that burden varies by transport system quality and housing-market structure. The Dhaka case is therefore not unique in experiencing housing–mobility trade-offs; what is distinctive is the combination of intense congestion, a large informal transport sector, and diverse household arrangements within a constrained housing market.

Taken together, the Bangladesh and regional literature makes three points very clearly. First, affordability is a binding constraint. Second, residence and mobility are connected. Third, inequality shapes who can access which locations. Yet this literature is still limited in three ways. It tends to privilege either housing cost or residential location without fully modeling daily mobility. It rarely incorporates living arrangement as a formal behavioral choice. And it often represents accessibility with crude distance or generalized location proxies rather than utility-based measures that capture the attractiveness of different modes. These limitations create a strong rationale for the present study.

2.4 Global Evidence and Theoretical Foundations

2.4.1 Urban Economic Foundations of Residential Location

The earliest theoretical foundation for residential location choice comes from urban economics, especially the bid-rent tradition (Alonso, 1964; Brueckner & Fansler, 1983; Termote, 1971). In this framework, land users trade off accessibility to central

opportunities against land or housing consumption. Locations closer to the urban core or high-opportunity areas command higher prices because they reduce travel cost, while more peripheral locations offer cheaper land or larger housing at the cost of distance. Although these models were formulated in highly stylized monocentric settings, their central insight remains durable where residential location reflects an exchange between accessibility and housing cost.

Later work complicated this picture by recognizing polycentric urban structure, dual-earner households, differentiated preferences, and the endogeneity of transport options. Giuliano & Small (1993) showed that commuting patterns are not fully explained by urban structure alone, while Sultana (2006) argued that jobs–housing balance research must better account for dual-earner households and intra-household complexity. These contributions are especially relevant for Dhaka, where households often coordinate not one but several activity locations such as workplaces, educational institutions, and care-related destinations within one residential decision.

2.5 Accessibility Measurement Evolution

Accessibility is generally understood as the ease with which individuals can reach desired opportunities, including workplaces, educational institutions, healthcare, and daily services. Hansen (2007) was among the earliest scholars to formalize the link between accessibility and land use. Later, Handy & Niemeier (1997) and Geurs & van Wee (2004) showed that accessibility can be conceptualized in multiple ways, including cumulative-opportunity measures, gravity-based measures, time–space approaches, and utility-based measures. The literature therefore evolved from viewing access as a simple matter of physical proximity toward a more sophisticated understanding that combines land-use opportunities with transport system performance.

This evolution matters because residential location research is highly sensitive to how accessibility is measured. Simple distance-based indicators are transparent and often useful, but they can misrepresent access in cities where travel conditions vary sharply by mode or by corridor. Even a widely used planning measure such as PTAL is intentionally simple, relying primarily on walking distance to stops and service frequency. In contrast, utility-based measures recognize that access depends on how travelers actually evaluate alternatives in terms of time, cost, convenience, and available mode sets (Akbari & Habib, 2015; Geurs & van Wee, 2004; Handy & Niemeier, 1997).

For the present thesis, the shift toward utility-based accessibility is crucial. The expected maximum utility, or logsum, from a mode choice model captures the combined attractiveness of available modes and is therefore more behaviorally consistent than distance alone. In urban settings such as Dhaka, where the same home–destination separation can be experienced through walking, rickshaw, bus, metro, ride-hailing, motorcycle, or mixed-mode chains, a logsum measure provides a more realistic representation of accessibility than a single travel-time or Euclidean-distance metric.

2.5.1 Location Affordability Concepts

Affordability is another concept that has expanded over time. Traditional housing studies often operationalized affordability as a ratio of housing expenditure to household income. Although this approach remains common, Stone & Stone (2010) argued that affordability is better understood through a residual-income lens: a household is only truly able to “afford” a dwelling if sufficient income remains after essential housing expenditure to cover other necessary costs. This argument becomes especially important when transport is included. A household may satisfy a conventional rent-to-income threshold and still be effectively burdened once commuting cost is added.

More recent literature has therefore advanced the concept of location affordability. Howell et al. (2018) and Makarewicz et al. (2020) showed that housing expenditure and transport expenditure should be evaluated together because urban form and network conditions influence both. A central area may be expensive in rent but relatively economical in total living cost if it reduces commuting. By contrast, peripheral housing may appear affordable in rent terms but become costly after accounting for travel time, fuel, fares, transfers, and car or motorcycle dependence. This framework is highly relevant to Dhaka, where the financial and temporal burden of travel can be substantial even over moderate distances.

2.5.2 Residential Self-Selection

A further contribution of the international literature is the recognition of residential self-selection. Schwanen & Mokhtarian (2005) argued that the relationship between neighborhood form and travel behavior is partly mediated by preferences where people may choose neighborhoods that fit their travel attitudes or desired lifestyles. Rashidi & Mohammadian (2015) similarly emphasized that housing search and choice-set formation are behavioral processes rather than purely mechanical responses to price and distance. These insights are important, but the Dhaka case suggests that self-selection

should not be overstated. In constrained housing markets, households often choose among limited feasible options rather than unconstrained preferred ones. The present study therefore benefits from the self-selection literature while remaining attentive to affordability-driven constraint.

2.6 Dwelling Arrangement, Modal Accessibility and Distance to PAL

2.6.1 Dwelling Arrangement Choice

Dwelling arrangement is an important but often underrepresented dimension of residential choice. Many studies in Western contexts examine living arrangement mainly through whether young adults leave the parental home, remain co-resident, or establish independent households (Mulder & Clark, 2002). That literature is useful, especially for understanding life-course transitions, but it is not fully adequate for South Asian megacities. In Dhaka and similar urban contexts, individuals may live alone or with non-family members, live with family without children, live with family with children, or live in joint-family arrangements. These categories are not simply descriptive household labels; they imply different housing needs, cost-sharing possibilities, mobility requirements, and spatial preferences.

Families with children may prioritize school access, safety, and dwelling space. Joint families may require larger units and may tolerate greater residential distance if suitable housing is available or if family support systems are maintained. By contrast, single individuals or shared non-family households may accept smaller spaces in exchange for proximity to work, educational institutions, or mixed-use neighborhoods. The literature increasingly supports this interpretation. Hasnine et al. (2023) showed that living arrangement and residential distance are jointly structured by modal accessibility, while Hawkins et al. (2021) highlighted the importance of household-level dynamics in residential location modeling. These findings strongly support the current thesis emphasis on living arrangement as an endogenous choice variable rather than a background control.

2.6.2 Household Activity Anchors

Household structure matters because households often negotiate several activity anchors at once. An employed adult may value workplace access, a student may value university proximity, parents may value school quality and caregiving support, and older family members may value access to social networks or health facilities. Thus, residential

location is not chosen relative to one destination only; it is a compromise among multiple household members with unequal bargaining power and different daily schedules. In this sense, the present thesis concept of a Primary Activity Location (PAL) which is defined as the main workplace or educational institution is analytically useful because it captures the dominant anchor around which residential distance is organized while still recognizing broader household complexity.

This logic is especially relevant in Dhaka because transport frictions are high and households frequently include both students and earners. The number of students, number of employed members, presence of children, and presence of older adults may all change the feasible residential search space. A family with school-going children may choose a residence near schools or coaching centers even if it lengthens one adult's commute. A young individual sharing housing with non-family members may instead prioritize direct access to employment or educational center and use co-residence as a cost-sharing strategy. For this reason, living arrangement should be treated as behaviorally connected to both residential location and accessibility, not as a fixed household descriptor.

2.6.3 Modal Accessibility Burden

Modal accessibility refers to the extent to which available transport modes allow individuals to reach desired destinations in terms of time, cost, convenience, reliability, and transfer burden. In residential studies, it is important because the same physical separation between home and destination may generate very different effective burdens depending on the available modes. A household with access to a private vehicle, ride-hailing, reliable bus service, and metro may experience a far larger feasible residential choice set than a household dependent mainly on walking or irregular buses (Akbari & Habib, 2015; Yao & Wang, 2014).

In Dhaka this point is particularly sharp. The city's modal environment is mixed and unequal: walking, rickshaw, bus, motorcycle, metro, ride-hailing, and paratransit do not provide equivalent reach, reliability, or comfort. First-mile and last-mile conditions also vary substantially across neighborhoods. Therefore, accessibility is not reducible to geometric distance. A residence six kilometers from a workplace may be manageable for one household and highly burdensome for another depending on transfers, waiting time, fare affordability, safety, walking infrastructure, and congestion. This is why the current study's use of mode-choice-based logsum accessibility is methodologically justified.

2.6.4 Work-Education Accessibility

The international literature has long linked accessibility to employment with residential location. However, in rapidly urbanizing cities, educational institutions play a similarly powerful role in shaping where households live. This is especially true for younger adults and for families with children. The present study therefore appropriately broadens the usual home–work focus by defining PAL around workplace or educational location. This is a significant conceptual move in the Dhaka context because education-related mobility is central to the city’s daily travel demand and to family residential strategy.

By linking dwelling arrangement with distance from PAL and embedding modal accessibility into that relationship, the study speaks to a literature that is increasingly concerned with how long-term residence choice and daily mobility co-produce one another. The contribution is especially relevant for Dhaka, where shared accommodation, family co-residence, and multimodal dependence are common strategies for managing housing and mobility constraints.

2.7 Integrated Modelling Traditions

2.7.1 Random Utility Foundations and The Move Beyond Separate Models

Discrete choice modeling provides the core analytical foundation for modern transport and residential choice research. In random utility terms, observed utility is represented as a systematic component plus a random component, and individuals are assumed to choose the alternative with the highest utility (Ben-Akiva & Lerman, 1985; McFadden, 1974). This paradigm has been extremely influential because it allows location, mode, tenure, and other urban choices to be modeled as probabilistic but behaviorally interpretable decisions.

Early applications often modeled residential choice, workplace choice, vehicle ownership, and mode choice separately. This produced important insights, but it also built in a strong assumption of independence across decisions that are often made jointly in practice. Abraham & Hunt (1997) showed that home, workplace, and commuter mode choices can be nested within one behavioral framework for multi-worker households. Bhat & Guo (2007) later demonstrated that built environment characteristics and auto ownership interact deeply with residential choice, while Pinjari et al. (2011) modeled the broader continuum linking residential location, vehicle ownership, bicycle ownership and commute tour mode choice. Together, these studies make a compelling argument

that residential and mobility behavior should not be treated as separate domains.

2.7.2 Utility-Based Accessibility Integration

A further advance in the literature was the embedding of accessibility directly into residential-choice models. Akbari & Habib (2015) showed that modal accessibility helps explain the home–work spatial configuration of two-commuter households. Hasnine et al. (2019) refined the idea further by proposing a random-utility-based accessibility measure for transit that explicitly captures first-mile constraints. Most relevant of all, Hasnine et al. (2023) showed that modal accessibility can shape both living arrangement and residential distance jointly. These studies are particularly important for the present thesis because they show that the expected maximum utility from a mode choice model is not merely a transport indicator; it is a land-use and housing variable as well.

This insight is highly relevant for Dhaka. In a city where congestion, transfer friction, informal transport, and unequal service quality are all common, utility-consistent accessibility is more informative than straight-line distance or average travel time. Two households living at the same metric distance from their PAL can face very different real burdens if one has metro access, safe walking conditions, and affordable feeder options while the other does not. Thus, using logsum accessibility allows the residential model to incorporate the actual opportunity set created by the transport system rather than a simplified proxy.

Evolution of literature informing the present study

From accessibility and location theory toward joint housing-mobility modelling in Dhaka

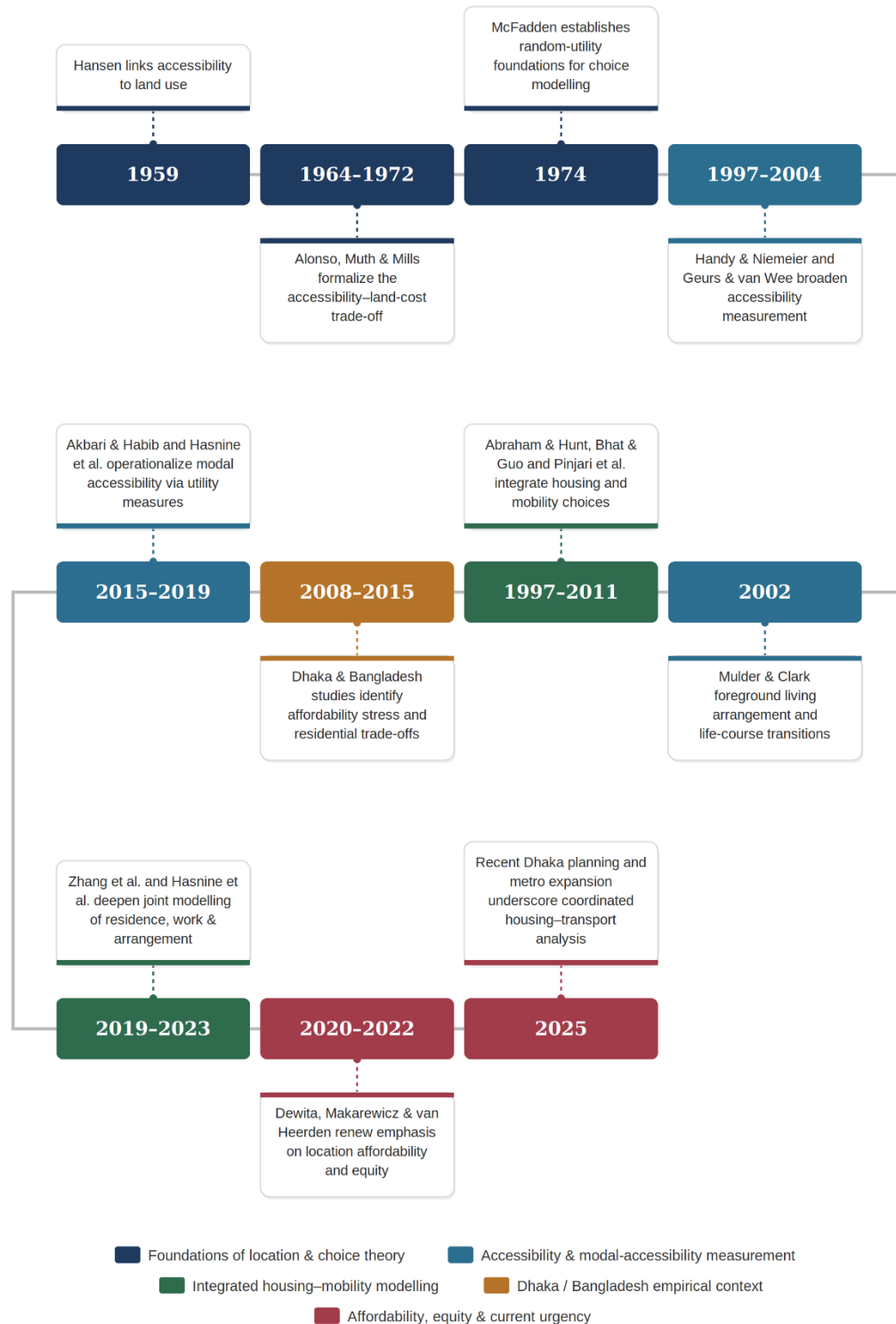


Figure 2.1: Evolution of literature informing the present study from accessibility and location theory toward joint housing mobility modeling in Dhaka

2.7.3 Joint Discrete–Continuous Modeling

Joint discrete–continuous modeling is especially appropriate when one part of the decision is categorical and another is scalar. In the present thesis, dwelling arrangement is discrete, while residential distance from PAL is continuous. A joint framework allows the model to recognize that the type of living arrangement selected by a household and the distance it is willing or forced to live from PAL are not independent outcomes. Unobserved preferences or constraints may influence both simultaneously.

This is a major methodological advantage over separate equations. If, for example, a household’s unobserved tolerance for congestion, crowding, or shared living affects both its likelihood of co-residing with family and its willingness to locate farther from work or study, a separate model may miss or misattribute that relationship. Joint modeling can capture these correlations more directly. That is why the literature increasingly treats residential location, mobility, and household form as co-produced behavioral outcomes rather than isolated choices.

Recent accessibility scholarship continues to stress that transport access is unevenly distributed and that poor transit service can deepen spatial inequality if planning ignores accessibility gaps. While this newer work often focuses on suburban or low-density contexts, its broader implication is directly relevant to Dhaka: accessibility is not only an efficiency issue; it is an equity issue, and unequal accessibility can widen the separation between affordable housing and urban opportunity (van Heerden et al., 2022).

2.8 Research Gaps and Conceptual Framework

2.8.1 Comparative Summary of Major Studies

The following table synthesizes studies most directly relevant to this thesis and clarifies how the present research extends them

Study	Location	Data and method	Core finding	Relevance to current thesis
Kamruzzaman & Ogura (2008)	Dhaka	Housing-market and middle-income housing analysis	Middle-income households face mismatch between income, supply, and suitable housing options	Grounds the affordability problem in Dhaka’s formal housing market
Jahan & Kalam (2012)	Dhaka	Rental affordability	Middle-income households face	Shows affordability is a binding

Study	Location	Data and method	Core finding	Relevance to current thesis
		assessment	significant rental burden	residential constraint
Choudhury & Ayaz (2015)	Bangladesh	Residential location choice modeling	Households may live far due to heterogeneous trade-offs involving cost and access	Directly links housing location to accessibility and cost considerations
Dewita et al. (2020)	Indonesia	Empirical analysis of housing and transport affordability	Housing and transport costs must be evaluated jointly	Supports the location-affordability perspective
van Heerden et al. (2022)	Developing-country planning context	Spatial planning and equity analysis	Accessibility, affordability, and equity must be addressed together in long-term planning	Reinforces the policy significance of integrated analysis
Bhat & Guo (2007)	United States	Joint residential-choice and auto-ownership econometric model	Built environment and ownership are endogenous to residential choice	Demonstrates why separate housing and mobility models are incomplete
Abraham & Hunt (1997)	Multi-worker household context	Nested logit model of home, work, and mode	Home, workplace, and mode choices are behaviorally linked	Early foundation for integrated household-level modeling
Pinjari et al. (2011)	United States	Integrated choice continuum model	Residential location, auto ownership, bicycle ownership, and commute mode are jointly determined	Strong methodological precedent for simultaneous decision modeling
Akbari & Habib (2015)	Canadian metropolitan context	Econometric model using modal accessibility	Modal accessibility influences home–work spatial configuration	Justifies using logsum accessibility in residential analysis
Hasnine et al. (2019)	Urban transit context	RUM-based transit accessibility measure	Utility-based accessibility better captures first-mile constraints	Supports behaviorally grounded accessibility measurement
Zhang et al. (2019)	Metropolitan residence–work context	Joint model of place of residence and place of work	Residential and workplace locations are interdependent under data limitations	Strengthens the rationale for linked residence–PAL analysis
Hawkins et al.	Household	Household-level	Intra-household	Highlights the

Study	Location	Data and method	Core finding	Relevance to current thesis
(2021)	residential-choice context	latent auction approach	dynamics matter for location choice	importance of household-level heterogeneity
Hasnine et al. (2023)	North American metropolitan context	Joint model of living arrangement and distance	Modal accessibility shapes both living arrangement and how far households live	Most directly related international precedent for this thesis

2.8.2 Synthesis of the Research Gap

The abovementioned review reveals a substantial gap at the intersection of housing, household structure, and mobility in Dhaka.

Firstly, Bangladesh-based studies have documented housing affordability pressures and aspects of residential location choice, but they rarely integrate travel behavior and household living arrangements within a single behavioral framework. In other words, the local literature identifies the problem but does not yet model its full decision architecture.

Secondly, much of the international living-arrangement literature remains too narrow for South Asian megacities. It often centers on the binary transition between living with parents and living independently. That simplification is inadequate for Dhaka, where a meaningful empirical classification must distinguish at least shared non-family living, family living without minors, family living with minors, and joint-family or extended-family living. These categories imply different budget constraints, different space requirements, different care responsibilities, and potentially different tolerances for commuting burden.

Thirdly, accessibility remains under-measured in most South Asia-focused work. Distance and average travel time are informative, but they do not fully represent the lived opportunity structure created by Dhaka's mixed and unequal modal environment. A utility-based accessibility measure derived from the mode choice model is better aligned with actual behavior because it reflects the combined effect of time, cost, availability, and mode quality. This is especially valuable in Dhaka, where the distinction between physical distance and effective accessibility is often sharp.

Fourth and lastly, the present literature says too little about residential distance from Primary Activity Locations when affordability and household structure are both considered. Job access has been studied far more than the broader set of workplace and

educational anchors that structure urban residence in Dhaka. Yet in a city with large student populations, mixed work–study trajectories, and family-led location decisions, residence relative to workplace or educational PAL is central to understanding mobility burdens and housing compromise.

2.8.3 Conceptual Framework

The conceptual logic of the current study is shown below. It treats dwelling arrangement and residential distance from PAL as jointly shaped by socio-demographic characteristics, household composition, housing conditions, built environment, and modal accessibility. Mode choice yields the expected maximum utility, or logsum, which serves as a behaviorally coherent measure of accessibility and informs both living arrangement and residential distance decisions.

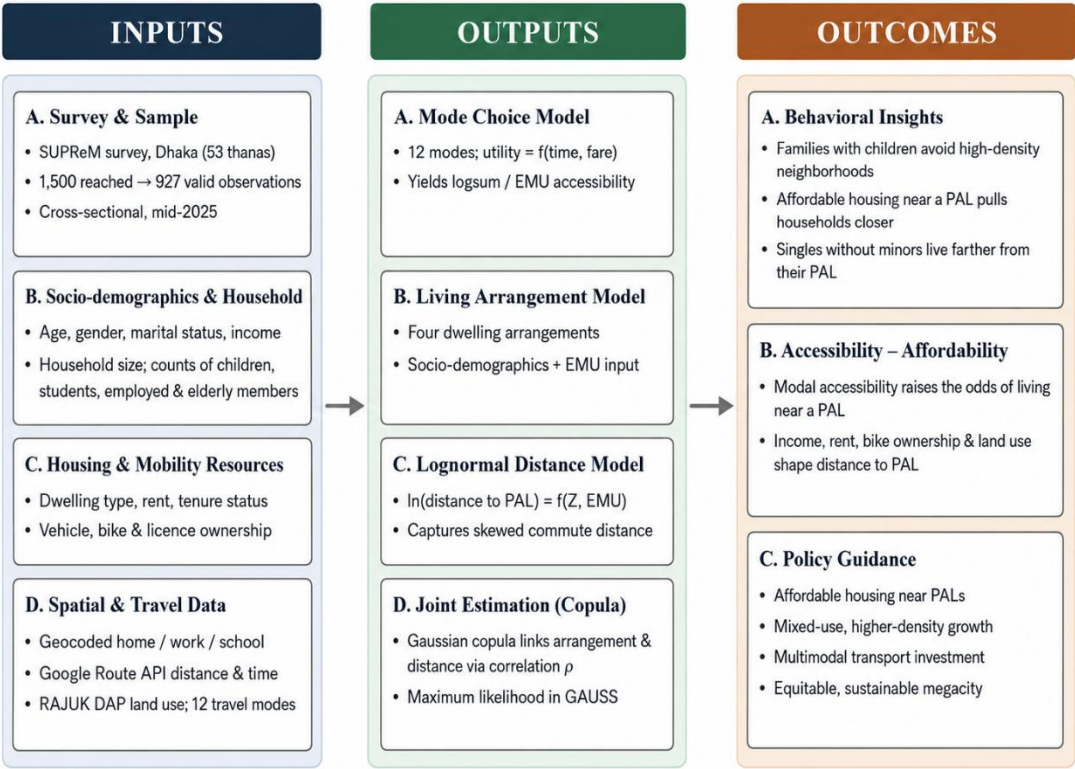


Figure 2.2: Logic-model framework of the joint dwelling-arrangement and mobility study, Dhaka

In summary, the literature suggests that urban households in rapidly growing and congested megacities negotiate a triad of constraints following where they can afford to live, how they can reach their primary activity locations, and with whom they can

realistically maintain a household. Dhaka intensifies this triad because of its dense and rapidly changing land use, its multimodal yet friction-filled transport environment, and its diverse household structures. The present thesis therefore occupies a distinct place in the literature by jointly modeling dwelling arrangement, mobility, and residential distance under affordability and accessibility constraints in Dhaka.

CHAPTER 3

METHODOLOGY

This chapter presents the methodological framework adopted to examine the interrelationship among dwelling arrangement, modal accessibility and residential distance from Primary Activity Locations (PALs) in Dhaka city. The analysis is structured around three connected components: the survey-based construction of individual and household-level data, the preparation of spatial and travel-related variables, and the formulation of a joint econometric model. The methodological approach is designed to capture the way individuals make residential and mobility-related decisions under affordability and accessibility constraints. First, the chapter describes the survey design, data collection procedure and preprocessing steps used to obtain the final analytical sample. It then presents the descriptive characteristics of the respondents, households, housing conditions and travel-related variables. Finally, the model formation is discussed, including the mode choice component, the living arrangement choice component and the lognormal distance component. Together, these elements provide the empirical basis for evaluating how socio-demographic characteristics, household structure, built environment conditions and modal accessibility influence residential decision-making in Dhaka.

3.1 Survey Design, Data Collection and Data Processing

The study utilized data from a household-level mobility and residential choice survey conducted in Dhaka, Bangladesh. The survey was broader than the present thesis and contained several modules suitable for different transportation and residential behavior analyses. It included revealed information on respondents' socio-demographic and household characteristics, typical travel behavior, residential and activity locations, vehicle availability, housing characteristics, attitudinal responses and stated preference choice tasks. However, the present thesis uses only the components relevant to the joint modeling of transportation mode choice, household living arrangement and residential distance from Primary Activity Locations (PALs). The stated preference component was not used in the current model. Similarly, attitudinal variables were used only to support descriptive interpretation and were not included as explanatory variables in the final joint econometric framework.

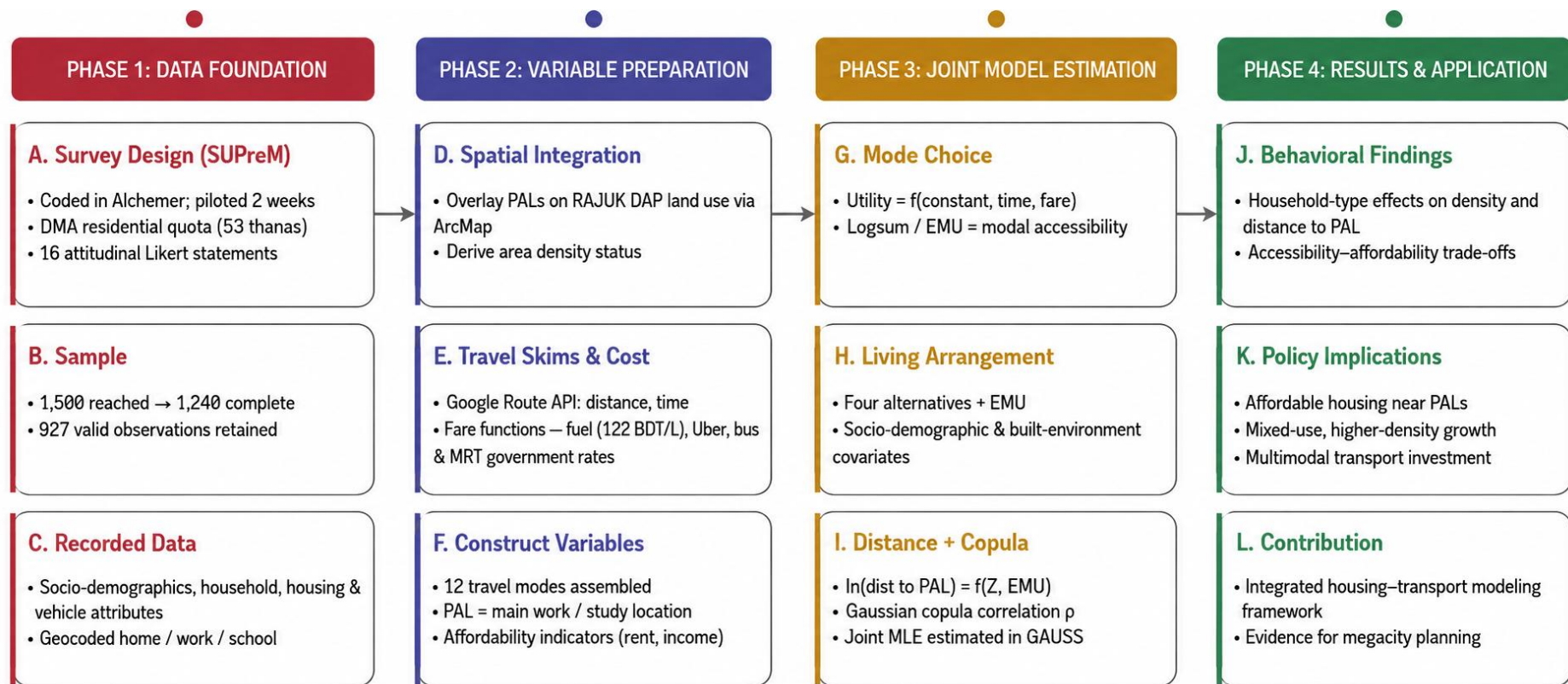


Figure 3.1: Methodological framework of the joint discrete–continuous model of dwelling arrangement and mobility

The final analytical dataset used in this thesis contains 927 valid observations. These observations were retained after applying multiple screening and preprocessing steps, including completeness checks, location validation, household consistency checks, vehicle access validation, driving license validation, timestamp-based quality control and route-based travel attribute generation. The following subsections describe the survey design, data collection process, validation framework, spatial processing, travel attribute preparation and sample calibration procedure.

3.1.1 Purpose and Conceptual Scope of the Survey

The main purpose of the survey was to construct a detailed dataset capable of explaining how individuals in Dhaka make interrelated decisions regarding residential location, living arrangement and travel to major daily activity locations. In dense and rapidly growing cities, the decision of where to live is not simply a housing decision. It is shaped by the interaction of housing affordability, access to work or education, family composition, vehicle availability, travel cost, travel time and modal accessibility. This is particularly important in Dhaka, where congestion and uneven transport service quality mean that physical distance alone does not fully represent the real travel burden experienced by residents.

The survey was therefore designed to capture the following dimensions of urban residential and mobility behavior:

- individual socio-demographic characteristics, such as age, gender, marital status, education, employment status, student status and personal income
- household characteristics, such as household size, number of students, number of employed members, number of children and number of elderly members
- housing and dwelling characteristics, including tenure type, rent level, dwelling type and household living arrangement
- mobility resources, including driving license ownership, access to private vehicle, motorcycle and bicycle availability
- spatial information, including geocoded residential location, workplace location and educational institution location
- travel behavior information, including most frequently used commute mode and relevant alternative modes
- residential preference and attitudinal information, including perceived importance of accessibility, safety, cost of living and neighborhood facilities.

These variables were selected because they collectively represent the major constraints and opportunities that influence residential and travel behavior. Socio-demographic characteristics capture differences among respondents, household variables represent family-level constraints, housing variables capture affordability and dwelling conditions, and travel variables represent the mobility environment. The spatial information provides the foundation for calculating distance and route-based travel attributes between residence and PAL.

In this thesis, PAL refers to the primary workplace or educational institution where a respondent spends the major portion of regular weekly activity time. The use of PAL allows the analysis to include both employed individuals and students. For employed respondents, the workplace was treated as the PAL. For students, the educational institution was considered the PAL. For respondents involved in both employment and education, the dominant regular activity location was identified based on their usual activity pattern. This definition is important because Dhaka's residential decisions are not only work-oriented; educational locations also strongly influence residential choice for students and households with education-related responsibilities.

3.1.2 Study Area and Sampling Frame

The survey was conducted in Dhaka, Bangladesh, covering both Dhaka North City Corporation and Dhaka South City Corporation. The study area includes 53 thanas, with 28 thanas in Dhaka North City Corporation and 25 thanas in Dhaka South City Corporation. A residential quota-based sampling approach was followed to maintain proportional representation across these thanas. Thana-wise population information was obtained from the relative census and used as the basis for determining the required distribution of responses across the study area (*Population-and-Housing-Census*).

The target population consisted of individuals aged 16 years or above who lived within the Dhaka Metropolitan Area. Since the objective was to analyze residential and mobility behavior within Dhaka, respondents outside the study area were not included. Screening questions were placed at the beginning of the survey to verify respondent eligibility and maintain the quota structure.

The screening process was used to determine whether:

- the respondent was at least 16 years old
- the respondent lived within the defined study area

- the respondent's residential thana belonged to the survey coverage area
- the quota for that residential thana was still open

If any of these conditions were not satisfied, the survey was terminated at the screening stage. This approach helped prevent invalid responses and ensured that the survey dataset remained aligned with the intended spatial sampling design. The use of thana-wise quotas was particularly important because Dhaka is highly heterogeneous in terms of land use, population density, transport accessibility and housing conditions. Without quota control, responses could have become concentrated in easily accessible or highly responsive areas, reducing the geographic representativeness of the dataset.

3.1.3 Survey Platform and Programming Logic

The survey was programmed in Alchemer, a web-based survey platform that supports questionnaire design, online survey administration, conditional branching, quota control and response monitoring (*Alchemer | Online Survey Software & Tools*). The use of a web-based platform was important because the questionnaire contained several linked sections and required logical conditions to ensure that respondents received relevant questions based on their socio-demographic and travel characteristics.

The platform was used to implement the following functions:

- informed consent and eligibility screening
- thana-wise quota control
- conditional display of questions based on employment or student status
- conditional display of vehicle-related questions based on access and availability
- map-based location collection
- timestamp recording at different stages of the questionnaire
- surveyor identification and response monitoring
- randomized display of stated preference tasks in the broader survey

The conditional logic in the questionnaire was useful for this thesis because it reduced irrelevant responses and helped identify internal inconsistencies during data cleaning.

For example, workplace location questions were shown only to respondents for whom employment-related travel was relevant. Educational institution location questions were asked where academic activity was relevant. Similarly, vehicle access questions were connected to household vehicle ownership questions. This structure improved response consistency and allowed later validation checks between related variables.

3.1.4 Questionnaire Organization

The questionnaire was organized into several sections. The sequence of questions was designed to begin with basic eligibility and socio-demographic information before moving to more detailed household, housing, travel and attitudinal questions. This structure was adopted to reduce early respondent fatigue and improve response quality. More sensitive or detailed household questions were placed later in the survey after respondents had already become familiar with the questionnaire.

The major sections of the questionnaire were:

- consent and screening
- personal and socio-demographic information
- household and dwelling information
- residential and PAL location information
- typical travel behavior
- residential preference and attitudinal statements
- stated preference choice tasks for the broader survey

Only the revealed and observed components relevant to the joint model were used in the present thesis.

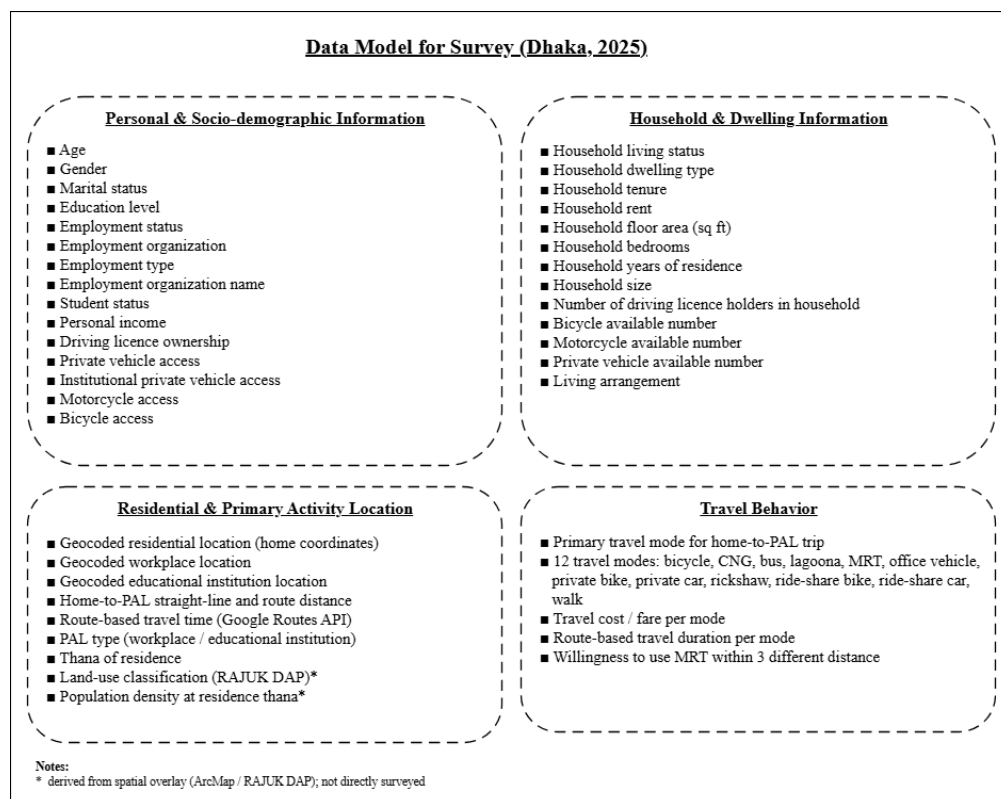


Figure 3.2: Survey data model of the survey questionnaire

3.1.4.1 Consent and screening section

The survey began with an informed consent statement. The statement explained the purpose of the study, the expected duration of the questionnaire, the voluntary nature of participation, and the general scope of the information being collected. Respondents were allowed to proceed only after providing consent.

After consent, two main screening questions were asked. The first screening question recorded the respondent's age, while the second recorded the respondent's residential thana. These questions were used to determine eligibility and quota status. Respondents below the age threshold, respondents living outside the DMA and respondents from thanas whose quotas were already fulfilled were screened out. This screening process helped ensure that the final dataset remained consistent with the study design.

3.1.4.2 Personal and Socio-Demographic Information

The personal information section collected individual-level variables that are commonly used in travel behavior and residential choice models. These included age, gender, marital status, education level, employment status, student status, personal income and driving license ownership. Such variables are important because residential location and travel mode choice are influenced by life-cycle stage, economic capacity, household responsibilities and mobility resources (Bhat & Guo, 2007; Pinjari et al., 2011).

In the context of Dhaka, marital status and student or employment status are especially relevant. Marital status is strongly connected with household living arrangement, while employment and student status determine the respondent's regular activity anchor. Education and income influence both housing affordability and modal choice, while driving license ownership affects the feasibility of using private motorized modes. Therefore, these variables were retained as important inputs for descriptive analysis and model development.

3.1.4.3 Household and Dwelling Information

The household section collected information on household structure, housing condition and dwelling arrangement. These variables were particularly important for the living arrangement component of the joint model. The survey recorded household size, number of students, number of employed persons, number of household members below 18 years, number of members aged 59 years or above, total household income, monthly rent, housing tenure, dwelling type and household vehicle availability.

The survey originally asked respondents to identify their household living arrangement using five categories:

- one person living alone
- one person living with non-family members
- family with no children below 18 years
- family with one or more children below 18 years
- joint family

For the final model, these responses were reorganized into four living arrangement alternatives:

- living separately
- family without minors
- family with minors
- joint family

The “living separately” alternative combines individuals living alone and those living with non-family members. This grouping was adopted because both categories represent non-family-based residential arrangements and because the number of respondents living with non-family members was relatively small. The other three categories were retained because they represent distinct household structures with different housing needs, mobility constraints and affordability considerations.

The distinction between family without minors and family with minors is behaviorally important. Families with minors may prioritize access to schools, safety, larger dwelling space and family-oriented neighborhood facilities. Joint families may have additional space needs and may accept different residential trade-offs due to the presence of elderly or extended family members. Therefore, living arrangement was treated not simply as a descriptive variable but as a behavioral choice outcome in the joint model.

3.1.4.4 Residential and Primary Activity Location Information

The survey collected geocoded residential location and activity location information using a map-based interface. Respondents were asked to provide the location of their current residence and, where applicable, the location of their workplace and educational institution. These coordinates were later used to calculate home-to-PAL distance and to generate route-based travel time and distance.

The geocoded location component was one of the most important parts of the survey for

the present thesis. Residential distance from PAL is the continuous outcome in the joint model, and therefore valid home and PAL coordinates were essential. Responses with missing, vague, inconsistent or spatially invalid location information were removed during data preprocessing.

PAL was defined as the main workplace or educational institution where the respondent spends most of the regular weekly activity time. This definition allowed the model to reflect the major spatial anchor of the respondent's daily life. In Dhaka, where travel conditions vary considerably by corridor and mode, a residence located the same physical distance from PAL may involve very different travel burdens depending on mode availability, congestion and transfer requirements. Therefore, the geocoded home-PAL relationship was necessary for constructing both distance and accessibility measures.

3.1.4.5 Typical Travel Behavior

The typical travel behavior section collected information on the respondent's most frequently used travel mode for the home-to-PAL trip. Respondents were also asked about other relevant travel modes and their usual travel pattern. The survey considered 12 travel modes, representing the multimodal transport environment of Dhaka:

Bicycle, CNG, bus, lagoon, MRT, office vehicle, private bike, private car, rickshaw, ride-share bike, ride-share car and walk.

The inclusion of these modes reflects the reality of Dhaka's transport system, where formal public transport, semi-formal paratransit, ride-hailing, private modes, non-motorized modes and walking operate simultaneously. This broad mode set was necessary for estimating a meaningful mode choice model and deriving a utility-based accessibility measure.

The travel behavior data were used to estimate the transportation mode choice component of the joint framework. From this model, the expected maximum utility or logsum was obtained. Logsum measures are widely used in transportation research because they provide a behaviorally consistent measure of accessibility by capturing the combined attractiveness of available alternatives (Akbari & Habib, 2015; Ben-Akiva & Lerman, 1985; Geurs & van Wee, 2004; Hasnine et al., 2019). In this thesis, the logsum was used as the measure of modal accessibility and was incorporated into the living arrangement and distance components of the model.

3.1.4.6 Residential Preference and Attitudinal Statements

The survey included attitudinal and residential preference statements to understand how respondents evaluate different factors when choosing a residential location. Responses were collected using a five-point Likert scale. The attitudinal items covered factors such as proximity to workplace or educational institution, access to public transport, access to quality schools, market access, recreational facilities, cost of living, communication services, safety and security, and availability of open space.

These attitudinal questions were useful for understanding the broader preference structure of Dhaka residents. However, the present thesis does not use these variables in the joint model. They are used only in the descriptive statistics section to illustrate the relative importance respondents place on different residential location factors. This approach keeps the econometric model focused on observed variables while still acknowledging that residential location decisions are influenced by subjective preferences and perceived neighborhood quality. The decision to limit attitudinal variables to descriptive analysis also avoids overcomplicating the joint model with latent psychological constructs, which are outside the central scope of this thesis.

3.1.5 Pilot Survey and Refinement Of The Questionnaire

Before the final survey, a pilot survey was conducted for approximately two weeks. The pilot included respondents from diverse backgrounds, including students, professionals, teachers and non-technical respondents. Around 50 responses were collected during this stage. The purpose of the pilot was not only to test the technical performance of the web-based questionnaire but also to evaluate whether respondents could understand and complete the questions accurately.

The pilot survey was used to assess:

- whether the wording of questions was clear and understandable
- whether respondents could follow the sequence of the questionnaire
- whether the survey length was acceptable
- whether the screening and quota logic worked properly
- whether the map-based location input was understandable
- whether respondents could correctly identify their household living arrangement
- whether vehicle ownership questions were interpreted consistently
- whether the mode categories were sufficiently clear for Dhaka's travel context

After the pilot survey, approximately one week was devoted to reviewing the responses and incorporating feedback. Question wording was refined, unclear instructions were revised, and the survey flow was improved. Particular attention was given to residential location input, workplace and educational institution location input, household type, vehicle access, and travel mode questions because these variables were essential for the final analytical dataset.

The pilot survey also helped identify potential sources of inconsistency. For example, respondents could misunderstand the difference between access to a vehicle and ownership of a vehicle. Similarly, some respondents required clarification about household living arrangement categories. These issues were addressed before final survey administration. Thus, the pilot phase served as a critical instrument-testing and quality-improvement stage.

3.1.6 Final Survey Administration

The final survey was administered for approximately one month beginning in mid-March 2025. Data collection was conducted using a combination of web-based self-completion and surveyor-assisted responses. A professional survey firm with access to a consumer research panel assisted in the data collection process. Respondents were invited randomly and could complete the form independently or with support from trained surveyors.

Although the survey platform was online, surveyor assistance was important for three main reasons. First, some respondents required help understanding the map-based location interface. Second, surveyors helped ensure that respondents from different thanas were reached according to the quota requirement. Third, field support improved response quality by reducing confusion and helping respondents complete the questionnaire correctly.

A surveyor training process was conducted before final deployment. Initially, 20 surveyors were selected for training. A three-hour training session was held to familiarize them with the questionnaire structure, respondent eligibility criteria, thana quota system, consent procedure, survey ethics, location input and quality control rules. Based on training performance and commitment, 12 surveyors were shortlisted for the final data collection team.

Several measures were taken to improve the credibility and traceability of the survey:

- surveyors received structured training before data collection

- surveyors were provided with official identification
- survey materials were marked with institutional identification to improve respondent trust
- each surveyor was assigned a unique ID
- surveyor-level response patterns were monitored
- suspicious response patterns could be traced back to the responsible surveyor
- thana-wise progress was monitored during the survey period

The final survey reached 1,500 individuals. Among them, 1,240 respondents completed the questionnaire, giving a completion rate of approximately 82.7%. However, the present thesis required valid information on residential location, PAL location, living arrangement, household composition and travel mode. Therefore, completed responses were not automatically included in the final model. Responses with missing, no geolocation of residence, inconsistent or invalid information in key variables were removed during data cleaning. After screening and preprocessing, 927 valid observations were retained for final analysis.

3.1.7 Data Quality Control and Validation Framework

A rigorous data quality control procedure was applied after data collection. Since the present thesis uses a joint econometric framework, data quality is especially important. Errors in household composition could affect the living arrangement model, errors in location information could affect the distance model, and errors in vehicle availability could affect the mode choice component. Therefore, several layers of validation were applied before finalizing the analytical dataset.

The validation procedure included:

- timestamp validation
- vehicle access validation
- household size and composition validation
- household living arrangement validation
- driving license validation
- surveyor location tracking
- surveyor-level response monitoring
- attitudinal response distribution inspection

These checks were used to identify responses that were completed too quickly, internally

contradictory, spatially suspicious or logically inconsistent. Only responses that passed validation procedures and had required variables for model development were retained.

3.1.7.1 Timestamp Validation

Five separate timestamps were embedded throughout the survey to monitor respondent progress and completion time. These timestamps were placed at different points in the questionnaire to evaluate whether respondents spent a reasonable amount of time on each major section.

For each timestamp, the 10th percentile of completion time was calculated. Responses with completion times below the 10th percentile threshold were treated as unusually fast. Such responses were considered unreliable because extremely short response times suggest that the respondent may have skipped reading the questions, answered randomly or completed the survey without sufficient attention.

This validation method helped identify low-quality responses that might not be obvious from missing values alone. It was particularly important because the questionnaire included detailed sections on location, household composition and travel behavior, which require a minimum level of attention and time. Responses that failed the timestamp validation were discarded before further analysis.

3.1.7.2 Vehicle Access Validation

Vehicle access and vehicle availability were collected in separate parts of the questionnaire. In the initial section, respondents were asked whether they had access to private cars, private bikes or adult bicycles. Later, in the household information section, they were asked to report the number of such vehicles available in the household.

These two responses were cross-checked. A response was considered inconsistent if the reported access status did not match the later household vehicle count. For example, if a respondent reported having no access to a private car but later reported one or more private cars in the household, the response was flagged. Similarly, if the respondent reported access to a vehicle but later provided contradictory vehicle-count information, the response was reviewed and removed when the inconsistency could not be resolved.

This validation was necessary because vehicle access affects mode availability in the mode choice model. Private car, private bike and bicycle alternatives cannot be interpreted correctly unless the respondent's access to these modes is consistently recorded. Since the logsum accessibility measure is derived from the mode choice model,

vehicle access errors could also affect the accessibility variable used in the living arrangement and distance models.

3.1.7.3 Household Size and Composition Validation

Household size was validated against several household composition variables. The survey collected total household size, number of students, number of employed members, number of members below 18 years and number of members aged 59 years or above. These variables were checked for logical consistency.

A response was discarded if:

- the number of students exceeded total HH size
- the number of employed members exceeded total HH size
- the number of HH members below 18 years exceeded HH size
- the number of HH members aged 59 years or above exceeded HH size
- the combined number of members below 18 and above 59 exceeded HH size
- the reported household structure was impossible based on the HH count variables

This validation was essential because household composition is central to the living arrangement model. The distinction between family without minors, family with minors and joint family depends directly on the presence of children and elderly members. Inconsistent household composition data could lead to misclassification of the dependent variable and bias the model interpretation. Therefore, strict validation rules were applied.

3.1.7.4 Household Living Arrangement Validation

The survey asked respondents to report their household living arrangement using five categories. These categories were then checked against reported household composition. This validation ensured that the selected living arrangement was consistent with the number of children and elderly members reported in the household.

The validation rules were as follows:

- if a respondent selected “family with one or more children below 18 years” but reported zero members below 18 years, the response was discarded;
- if a respondent selected “joint family” but reported zero members aged 59 years or above, the response was discarded
- if a joint family also included children, the response was retained as a joint family
- for joint family validation, the elderly-member criterion was prioritized
- living alone and living with non-family members were checked against household

size and household relationship information where applicable.

This step was particularly important because living arrangement is one of the core dependent variables in the joint model. The purpose of the model is not merely to describe living arrangement but to explain how it is associated with accessibility, affordability, household structure and distance from PAL. Therefore, living arrangement categories had to be internally consistent before model estimation.

3.1.7.5 Driving License Validation

Driving license information was also validated using two related questions. In the first phase, respondents were asked whether they had a valid driving license. Later, they reported the total number of driving license holders in the household. These responses were compared to identify contradictions.

If a respondent reported having a driving license but later reported zero license holders in the household, the response was considered inconsistent and removed. This validation was necessary because license ownership is related to private motorized mode availability and can influence travel mode choice. Errors in license information could therefore affect the mode choice component and the derived accessibility measure.

3.1.7.6 Location Tracking and Surveyor-Level Validation

Surveyor location was tracked using geolocation data. The survey process included area codes, and the area code from the seven-digit survey code indicated the expected survey location. If the recorded surveyor location appeared suspicious or inconsistent with the intended area, the response was flagged. Responses were removed when the location record suggested that the survey was not conducted in the expected area.

Surveyor-level validation was also conducted using unique surveyor IDs. This allowed the dataset to be monitored by surveyor and helped identify unusual response patterns. For example, if a surveyor produced a large number of unusually fast responses, geographically inconsistent records or repeated patterns, those responses could be reviewed. This process improved the traceability and credibility of the survey data.

Location tracking was important for maintaining the reliability of the thana-wise quota design. Since the survey intended to represent different parts of Dhaka, inaccurate field location records could distort the spatial distribution of the sample. Therefore, geolocation-based quality checks were used as part of the final screening process.

3.1.7.7 Attitudinal Response Quality Inspection

The distributions of attitudinal responses were inspected to assess response variability and detect possible straight-lining. Straight-lining occurs when respondents choose the same response option repeatedly without carefully evaluating each statement. Although attitudinal variables were not included in the final joint model, their response patterns were still useful for judging overall response quality.

The attitudinal variables covered residential preference, accessibility, safety, cost of living and neighborhood-related considerations. Responses with suspiciously uniform patterns were reviewed together with other quality indicators such as timestamp values and surveyor records. This step helped ensure that the retained observations came from respondents who engaged meaningfully with the questionnaire.

3.1.8 Data Cleaning and Final Sample Formation

The raw completed survey dataset contained 1,240 responses. These responses were processed through several stages before preparing the final analytical sample. The first stage removed responses that did not satisfy eligibility criteria or had incomplete essential information. The second stage focused on location validity because the model required both residential location and PAL location. The third stage removed responses with contradictory household, vehicle or license information. The fourth stage prepared spatial, travel time, travel cost and land-use variables.

The final sample was formed using the following criteria

- the respondent had to be eligible based on age and study area
- residential location had to be valid and geocoded
- PAL location had to be valid and geocoded
- household living arrangement had to be internally consistent
- household size and composition variables had to pass logical checks
- vehicle access and household vehicle-count variables had to be consistent
- driving license information had to be consistent
- survey completion time had to pass timestamp-based validation
- surveyor location records had to be credible
- route-based travel attributes had to be computable

After applying these criteria, 927 valid observations were retained. This final dataset was used for descriptive analysis and for estimating the joint model. The reduction from 1,240

completed responses to 927 valid observations reflects the strict data requirements of the present modeling framework. Since the model jointly analyzes mode choice, living arrangement and distance from PAL, only observations with complete and consistent information across all three dimensions were retained.

3.1.9 Spatial Data Processing and Built-Environment Variable Preparation

Spatial data processing was conducted to link survey responses with built-environment information. The geocoded residential and PAL locations were integrated with RAJUK's Detailed Area Plan using ArcMap (*RAJUK Masterplan*, n.d.)(*ArcMap*, n.d.). This spatial overlay enabled the identification of land-use characteristics associated with respondents' residential locations and primary activity locations.

The spatial processing generated variables related to:

- residential land-use context
- mixed-use environment
- high-density or lower-density area status
- spatial relationship between residence and PAL
- built-environment characteristics around the respondent's location

These variables were included because land-use and density can influence both residential location and living arrangement. For example, mixed-use areas may offer better access to transport, markets and services, making them attractive for individuals who prioritize accessibility. However, high-density areas may also involve higher rents, smaller dwelling space, crowding and congestion, which may discourage larger households or joint families. Therefore, the built-environment variables provide an important connection between urban form and household residential behavior.

The use of RAJUK's Detailed Area Plan helped ensure that land-use classification was based on an official planning dataset. ArcMap was used to perform the spatial overlay and assign land-use attributes to each observation. This process transformed the survey dataset from a purely questionnaire-based dataset into a spatially enriched dataset suitable for integrated land use–transport analysis.

3.1.10 Route-Based Travel Distance and Duration Estimation

The survey did not collect complete travel diary data. Therefore, travel distance and duration between residence and PAL were estimated using the Google Routes API. The API was used by providing origin and destination coordinates and relevant travel mode

information (*Get a Route | Routes API | Google for Developers*). This approach yielded standardized route-based distance and duration measures for each respondent.

Using API-based travel attributes has several advantages. First, it reduces reliance on self-reported travel time, which may be affected by recall error. Second, it provides a consistent method for calculating distance and duration across respondents. Third, it allows travel attributes to be generated for multiple modes, which is necessary for estimating the mode choice model.

The route-based distance from home to PAL was also used to support the continuous distance component of the joint model. Because commuting distance is positive and often right-skewed, the distance model later assumes a lognormal distribution. The computed distance and duration variables therefore serve as key inputs in both descriptive analysis and model formation.

3.1.11 Travel Cost and Fare Variable Preparation

Travel cost variables were prepared for each relevant mode using mode-specific assumptions. Since Dhaka's transportation system includes formal, informal, private, public and ride-hailing modes, cost preparation required several different sources and assumptions. Travel cost was necessary because the mode choice component uses travel time and travel cost as key explanatory variables.

The cost preparation followed these general principles:

- for private car and private bike, operating cost was calculated using contextual fuel price and standard cost assumptions;
- the fuel price assumption was based on available contextual fuel price information and standard operating-cost references (i.e., 122 BDT/liter) (*Cost of Driving from DAC to Dhaka, Bangladesh; Finance News: Latest Financial News, Finance News Today in Bangladesh*)
- ride-share bike and ride-share car costs were estimated using Uber fare functions applicable to Bangladesh (*Earn Money by Driving or Get a Ride Now | Uber Bangladesh*).
- rickshaw fare was assigned based on local travel experience and prevailing fare practices
- bus and lagoon fares were initially set using the government fare basis;
- MRT fare was calculated using the government-approved rate of Tk. 5 per

kilometer (*Metro Rail Fare Fixed at Tk5 per Km*)

- walking and cycling were treated as zero-fare modes, though they still involve time cost through the travel-time variable

This mode-specific cost preparation allowed the model to compare different travel alternatives in a consistent utility framework. In Dhaka, travel cost is a major determinant of travel behavior, especially for low- and middle-income respondents. Therefore, incorporating travel cost into the mode choice component was necessary for generating a meaningful accessibility measure.

3.1.12 Final Analytical Dataset

The final analytical dataset was produced through a sequential process of survey design, pilot testing, final administration, response validation, spatial processing, route attribute generation, travel cost preparation and sample calibration. The raw survey reached 1,500 individuals and produced 1,240 completed responses. However, the present thesis required a stricter dataset as the joint model depends on valid information from several linked domains: living arrangement, PAL distance and transportation mode choice.

The final dataset includes:

- validated socio-demographic variables
- validated household composition variables
- validated housing and dwelling arrangement variables
- geocoded residential and PAL locations
- land-use and built-environment attributes from spatial overlay
- reported most frequently used travel mode
- route-based travel distance and duration
- mode-specific travel cost estimates
- derived modal accessibility measure from the mode choice model
- descriptive attitudinal information on residential preference

After all screening and preprocessing steps, 927 valid observations were retained. This final sample forms the basis of the descriptive analysis and joint model estimation presented in the following sections. The dataset is appropriate for analyzing how modal accessibility, affordability-related characteristics, household structure, built-environment conditions and mobility resources jointly influence living arrangement and residential distance from PAL in Dhaka.

3.2 Descriptive Analysis

The descriptive analysis of the survey data provides useful information about the composition of the study population, household structures, mobility resources, and the factors considered important when choosing a residential location.

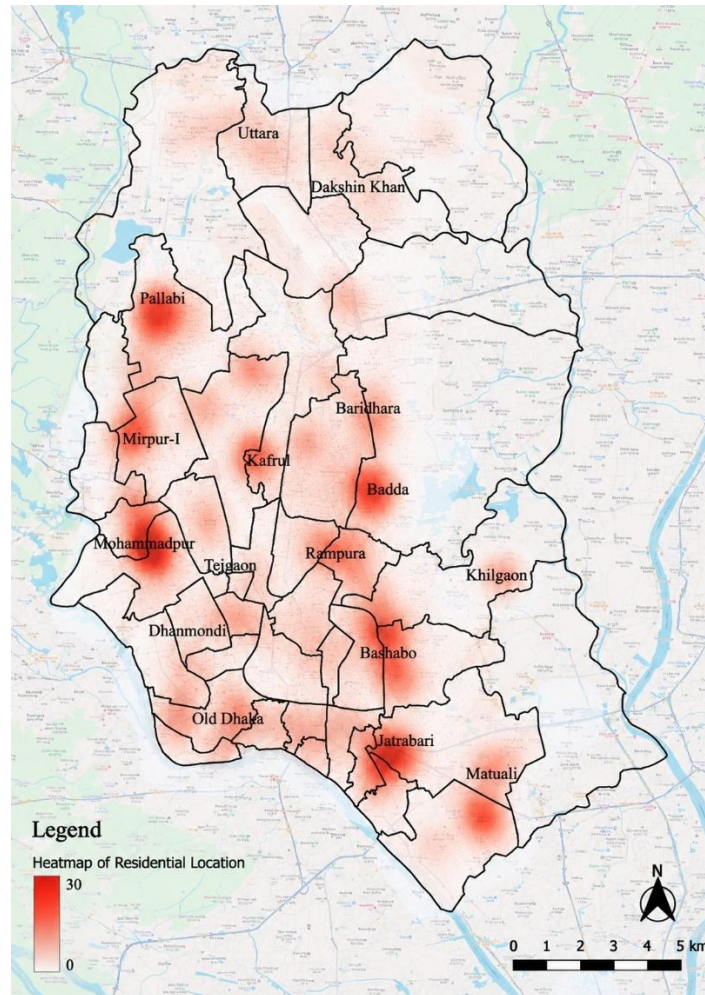


Figure 3.3: Spatial Heatmap of Respondents' Residential Locations

The sample is predominantly male, with men making up 71% of the participants and women making up 29%. A majority (60%) are married, while 38% are unmarried. This shows how important family is in urban society. Most of the people who attended the survey live in nuclear families, with 36% living in households without kids under 18 and 37% living in families with kids—an extra 17% live in joint families, which is still a big deal in South Asia. Only 8% of people live alone and 2% with non-family members, consistent with trends observed in other megacities, where cultural norms and economic realities determine how people live.

Housing conditions are characterized by a high proportion of multi-storied buildings

(51%) and rental tenure (73%), consistent with the constraints of Dhaka’s dense urban environment and a limited supply of affordable housing and house ownership opportunities. Only 19% of households are owner-occupied, underscoring a reliance on rental housing that reflects trends in high-density global cities.

Private car ownership is minimal, with merely 8% indicating access to a private vehicle and only 18% possessing a licensed household member. Ownership of bicycles (5%) and motorcycles (19%) is restricted. However, access to public transit and non-motorized means is extensive—this observation corresponds with the relatively short average distance to principal activity areas (mean = 7.6 km, SD = 5.8 km). The average home consists of approximately four individuals, with a mean age of 33 years, including both younger (mean number under 18 = 1.22) and elderly (mean above 59 = 0.46) generations, indicative of the continuation of intergenerational living arrangements.

Attitudinal responses reveal that urban resident in Dhaka place substantial weight on accessibility and safety when choosing their residential locations. Over 90% rate proximity to work or educational institutions, access to public transport, quality schools, and safety/security as important. Cost of living, market access, and recreational amenities are also key considerations, emphasizing a multi-dimensional view of residential desirability that combines practical, economic, and social factors.

Table 3.1: Summary statistics of respondent socio-demographic, household, and attitudinal characteristics

Discrete variables			
Variables	% Share	Variables	%Share
Gender		Household living status	
Male	71%	Living separately	10%
Female	29%	Family without a child under 18	36%
Marital status		Family with a child under 19	37%
Single	38%	Joint family	17%
Married	60%	Household dwelling type	
Highest level of education		Multi-storied building	51%
Higher education	46%	Mess/Hostel	6%
Up to HSC	54%	Other	43%
Student status		Household rent (in BDT)	
Full time	25%	Less than 10K	15%
Part time	2%	10K-30K	72%

Discrete variables			
Variables	% Share	Variables	%Share
None	73%	More than 30K	13%
Employment status		Household tenure type	
Employed	64%	Rented	73%
Unemployed	36%	Owned	19%
License ownership		Other	8%
Yes	18%	Land use type	
No	82%	Residential	45%
Access to private vehicle		Mixed	27%
Yes	8%	Other	28%
No	92%	Area density status	
Access to bike		High density	45%
Yes	19%	Low density	55%
No	80%	Personal income per month (in BDT)	
Family income per month (in BDT)		Less than 25K	19%
less than 25K	8%	25K-50K	24%
25K-50K	32%	More than 50K	57%
75K-100K	35%		
More than 100K	25%		
Continuous variables			
Variables	Mean	St. Dev.	
Distance from home to primary activity location (km)	7.59	5.79	
Respondent age	33.1	9.55	
Household size	4.01	1.39	
Number of license holders in the household	0.33	0.60	
Number of students in the household	1.36	0.64	
Number of employed persons in household	1.026	0.75	
Number of members below 18 years in household	1.22	0.76	
Number of members above 59 years in household	0.46	0.68	
Number of available private vehicle in household	0.10	0.38	
Number of available bikes in household	0.21	0.45	
Number of available bicycles in household	0.03	0.20	
Attitudinal response			
Importance of Factors in Residential Location Choice			
Conditions	% Less important	% Neutral	% Important
Proximity to one's own workplace	3%	6%	91%
Proximity to family members' workplace or educational institute	2%	4%	94%

Discrete variables			
Variables	% Share	Variables	%Share
Quality of schools/educational institutions nearby	2%	3%	95%
Access to public transport	3%	4%	92%
Access to the Market or supermarket	10%	4%	86%
Access to recreational facilities (parks, gyms, etc.)	8%	5%	86%
Cost of living	1%	4%	95%
Fast and cheap Internet and communication service	9%	5%	86%
Safety and security of the neighborhood	1%	2%	96%
Open space nearby to get enough air and sunlight	5%	4%	91%

3.3 Model Formation

In this study, a sophisticated joint model approach is employed using GAUSS programming software (Brook-Hart & Jakeman, 2012) to simultaneously analyze the interrelated choices of living arrangement, commute distance, and transportation mode choice. The primary objective of this modeling is to capture the complex interdependencies and correlations of individuals' decision-making process using an integrated Multinomial Logit-Normal (MNL-Log Normal) framework. This complicated method of discrete choice modeling makes it possible to capture heterogeneity and correlation between different behavioral components, which provides a better understanding of people's decision-making process. The joint model structure includes three interconnected decision-making modules:

- Transportation mode choice
- Living arrangement choice
- Commuting distance (Assumed log normal distribution)

These choices are modeled jointly to illustrate how socio-demographic factors, built environment characteristics, and travel-related variables influence people's decisions.

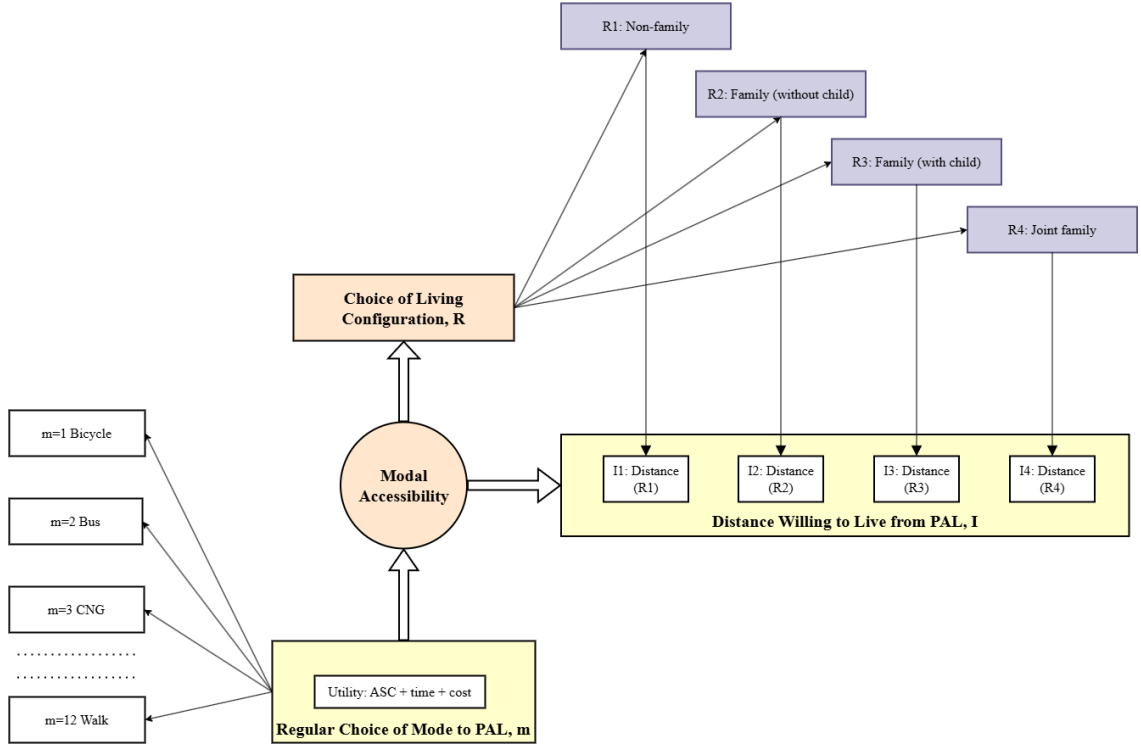


Figure 3.4: Conceptual framework of the joint location–mode choice model, illustrating the interdependence among living arrangement, modal accessibility, and distance from Primary Activity Locations (PALs).

The transportation mode choice model examines 12 different types of travel modes described before. Each model alternative's systematic utility is based on constant, travel time, and travel fare/cost. The functional form of mode choice utility:

$$U_{M,im} = V_{M,im} + \epsilon_m \quad (1)$$

$$V_{M,im} = \delta_m + \zeta_m T_{im} + \eta_m F_{im} \quad (2)$$

Where δ_m is the alternative-specific constant, ζ_m and η_m represent coefficients for travel duration T_m and fare F_m respectively for mode m .

The probability of choosing mode m is:

$$P_{M,im} = \frac{\exp(V_{M,im})}{\sum_{m'} \exp(V_{M,im'})} \quad (3)$$

The expected maximum utility (EMU) or logsum for the mode choice model is:

$$EMU_i = \ln(\sum_{m'} \exp(V_{M,im'})) \quad (4)$$

The logsum is a theoretically consistent, utility-based measure of accessibility that captures the combined attractiveness and availability of all transportation alternatives

between an individual's home and primary activity location. This accessibility index (EMU) is then included as an explanatory variable in the other model components.

The living arrangement choice components are categorized into four distinct alternatives according to individual living conditions: living separately (alone or with non-family members), family without children, family with children, and joint family. For each alternative, utility functions are specified, incorporating relative explanatory variables such as marital status, gender, number of students in the household, respondent's education level, vehicle accessibility, housing tenure type, household size, individual personal income, and built environment indicators. Accessibility measure is included through the logsum of the mode choice model.

The utility function for living arrangement choice $U_{LA,i}$ is specified as follows:

$$U_{LA,iR} = V_{LA,iR} + \epsilon_R \quad (5)$$

$$V_{LA,iR} = \alpha_R + \sum_j \beta_{jR} X_{ij} + \gamma_R \cdot EMU_i \quad (6)$$

Here, $V_{LA,iR}$ denotes the utility associated with the living arrangement alternative i , α_R is the alternative-specific constant, β_{jR} are the parameters for explanatory variables X_{ij} , γ_R represents the parameter associated with the logsum (LS), and ϵ_R represents the error term. The error term is assumed independently and identically distributed (IID) following a Gumbel distribution.

For the distance model, the Primary Activity Location (PAL) is defined as the location where individuals actively spend most of their weekly hours in either academic or work-related activity. The commuting distance component is modeled using a log normal regression approach, allowing commuting distance to vary systematically according to socio-economic factors, land use types, household characteristics, vehicle ownership, and environmental factors. The model assumes that the commute distance follows log normal distribution. The logged distance is modelled as follows

$$\ln(l_i) = \theta_R + \sum_h \phi_{hR} Z_{ih} + \lambda_R \cdot EMU_i + \xi_i \quad (7)$$

Here, θ_R is the intercept, ϕ_{hR} are parameters associated with explanatory variables Z_{ih} , and ξ_j is the normally distributed error term.

The probability density for observing distance l_i (conditional on arrangement R) is:

$$f_{l,iR}(l_i) = \frac{1}{l_i \sigma \sqrt{2\pi}} \exp\left(-\frac{[\ln(l_i) - \mu_{iR}]^2}{2\sigma^2}\right) \quad (8)$$

Where $\mu_{iR} = \theta_R + \sum_h \phi_{hR} Z_{ih} + \lambda_R \cdot EMU_i$.

Crucially, the correlation between living arrangement choice and commuting distance is captured via the correlation parameter. A standard normal cumulative distribution function (CDF) transformation is applied to the probability of each living arrangement, which produces a cumulative probability distribution. When multiplied by probability density function (PDF) for commuting distance, these distributions jointly yield the joint probability of selecting a specific living arrangement and associated commuting distance. The joint probability is mathematically represented as:

$$P_{joint,iR}(l_i) = f_{l,iR}(l_i) \cdot \Phi_2\left(\Phi^{-1}(P_{LA,iR}), \frac{\ln(l_i) - \mu_{iR}}{\sigma}; \rho\right) \quad (9)$$

Where P_{LA} is the probability of selecting a living arrangement, Φ and Φ^{-1} denote the CDF and inverse CDF of the standard normal distribution, μ_D and σ_D are the mean and standard deviation of the logged commuting distance, and ρ represents the correlation parameter.

For each individual i , let δ_{iR} and δ_{im} be indicator variables for the observed living arrangement and commuting mode.

The likelihood for an individual i is:

$$L_i = [P_{joint,iR}(l_i)]^{\delta_{iR}} \cdot [P_{M,im}]^{\delta_{im}} \quad (10)$$

The full log-likelihood is:

$$\mathcal{L} = \sum_i (\sum_R \delta_{iR} \ln P_{joint,iR}(l_i) + \sum_m \delta_{im} \ln P_{M,im}) \quad (11)$$

The model parameters are estimated using maximum likelihood estimation (MLE) through GAUSS. The methodological framework effectively captures the nuanced interdependencies among living arrangements, mode choice, and commuting distance.

3.4 Summary of Methodological Scope

This methodology provides an integrated framework for analyzing the joint relationship among transportation mode choice, dwelling arrangement choice and residential distance from Primary Activity Locations. The framework begins with the construction of a survey-based dataset that combines socio-demographic characteristics, household

attributes, housing conditions, land-use information, vehicle availability, travel mode characteristics and geocoded residential and activity locations. This allows the analysis to represent the individual and household dimensions of residential decision-making.

The methodological structure is developed in a sequential but behaviorally connected manner. The mode choice component first estimates the utility associated with available travel alternatives using travel time and travel cost. From this component, the expected maximum utility, or logsum, is derived as a measure of modal accessibility. This accessibility measure is then incorporated into the living arrangement and distance components of the model. By doing so, the model does not treat accessibility as a simple physical measure of distance; rather, it represents the relative attractiveness of transport opportunities available to each respondent.

The living arrangement model captures the discrete choice among different dwelling configurations, including living separate, living with family without children, living with family with children and living in a joint family. These alternatives reflect the household structures commonly observed in Dhaka and allow the model to examine how family composition, income, housing tenure, dwelling type and transport accessibility influence residential arrangements. The distance model, in contrast, treats the residential distance from PAL as a continuous outcome and assumes a lognormal distribution to reflect the positive and skewed nature of commute distance.

A key feature of the methodology is the joint treatment of living arrangement and residential distance. Instead of estimating these outcomes independently, the adopted framework allows the model to account for correlation between unobserved factors affecting household configuration and willingness to live at a certain distance from the main workplace or educational location. This is important because residential decisions are rarely made in isolation. Household needs, affordability constraints, access to transport and activity locations jointly influence both who individuals live with and how far they live from their PAL.

Overall, the methodological scope of this study is limited to the analysis of observed residential and mobility behavior using cross-sectional survey data. It does not attempt to model long-term residential relocation dynamics, housing supply mechanisms or changes in land price over time. Nevertheless, the framework provides a suitable basis for identifying the behavioral links among accessibility, affordability, household

structure and residential distance in a dense urban context. The results derived from this methodology are expected to support a more integrated understanding of housing and transportation planning issues in Dhaka city.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Overview

This chapter presents the empirical results of the joint econometric model developed in CHAPTER 3 and discusses their behavioral and policy implications. The model simultaneously estimates three interrelated choices of urban residents in Dhaka: the regular choice of commuting mode to the Primary Activity Location (PAL), the discrete choice of living arrangement, and the continuous commute distance between the residence and the PAL. The three components are connected in two ways. First, the expected maximum utility (EMU), or logsum, derived from the mode choice component enters the living arrangement and distance components as an endogenous measure of modal accessibility (Akbari & Habib, 2015; Hasnine et al., 2019). Second, the living arrangement and distance components are linked through a Gaussian copula that captures the correlation between their unobserved utility components (Hasnine et al., 2023). The joint structure therefore allows the analysis to answer not only who lives with whom and how far people live from their PAL, but also how the perceived performance of the transport system mediates the trade-off between these two long-term decisions.

The chapter is organized as follows. Section 4.2 presents the estimation results of the joint model, beginning with the model specification strategy and goodness-of-fit statistics, followed by a detailed interpretation of the mode choice component, the dependence structure between living arrangement and commute distance, the living arrangement choice component, and the commute distance component. Section 4.3 presents the marginal effects computed from the living arrangement model, organized by alternative, while Section 4.4 presents the marginal effects of the commute distance model for each living arrangement segment. Section 4.5 translates the empirical findings into policy recommendations for housing, transportation, and land-use planning in Dhaka, and Section 4.6 summarizes the chapter.

Throughout the chapter, parameter estimates are evaluated at a 95 percent confidence level together with their expected behavioral signs. Following the practice of comparable joint modeling studies (Hasnine et al., 2023; Pinjari et al., 2011), a number of parameters with lower statistical significance are deliberately retained in the final specification

because they carry behavioral insight, stabilize the specification of the corresponding utility or distance equations, and allow the marginal effect computations of Sections 4.3 and 4.4 to reflect the complete behavioral structure rather than only the strongly significant subset of effects

4.2 Model Estimation

4.2.1 Specification Strategy and Goodness of Fit

The final joint model is estimated on 927 valid observations drawn from the survey described in CHAPTER 3, using full-information maximum likelihood programmed in GAUSS with the MAXLIK estimation routine. The specification was developed iteratively: an extensive candidate set of socio-demographic, household, housing, mobility, and built-environment variables was first introduced in each component, and the specification was then progressively reduced by removing variables that were both statistically insignificant and behaviorally uninformative, while checking after every reduction that the signs and magnitudes of the remaining parameters were stable. Particular care was taken to avoid endogeneity in the living arrangement component: distance-based variables were excluded from the living arrangement utilities, since commute distance is itself an outcome of the joint structure, and any correlation between the two choices is instead absorbed by the copula dependence parameter.

Table 2 reports the estimation results. The final model contains 111 estimated parameters across the three components, together with the scale parameter of the lognormal distance distribution and the copula correlation coefficient. The log-likelihood of the full model is -677.71 against a null log-likelihood of $-4,698.67$, which yields a rho-squared value of 0.86 and an adjusted rho-squared value of 0.83. Because the null benchmark of equally likely alternatives is a weak standard for models with many alternatives, the fit was also evaluated against a constants-only benchmark, against which the model attains a rho-squared value of 0.74 and an adjusted value of 0.69. The substantial improvement over the constants-only benchmark indicates that the explanatory variables, rather than merely the observed market shares, account for most of the explained variation. Considering the complexity of the model, which jointly reproduces a 12-alternative mode choice, a 4-alternative living arrangement choice, and a segmented continuous distance distribution within a single likelihood, this constitutes a very good statistical fit (Ben-Akiva & Lerman, 1985; McFadden, 1974)

Table 4.1: Estimation of parameters of the joint model

Fit Indices:	
1. Loglikelihood of full model	677.71
2. Loglikelihood of null model	-4698.67
3. Parameters estimated	111
4. Rho squared value	0.86
5. Adjusted rho squared value	0.83
6. Constant only rho squared value	0.74
7. Adjusted constant only rho square value	0.69

Mode Choice Model Estimations				
		Constant	Duration	Fare
Public Transit	Bus	1.28***		
	MRT	1.37***		
Paratransit	CNG	3.31***		
	Office Vehicle	1.20**		
	Ride Share Bike	-0.03 (0.98)		
	Ride Share Car	4.60***	-0.112***	-0.034***
	Rickshaw	2.27***		
	Lagoona	-0.502 (-1.21)		
Private	Bicycle	<i>ref</i>		
	Private Bike	1.87***		
	Private Car	3.09***		
	Walk	0.87*		

Discrete Choice Model: Living Arrangement Choice				
Variable	Non-Family (reference)	Family, No Child	Family, With Child	Joint Family
Constant	<i>ref.</i>	-26.19**	-16.77	-35.77***
<i>Individual characteristics</i>				
Log of age		8.69**	7.06**	11.02***
Gender: male				-0.43 (-0.64)
Marital status: single		2.73***		3.40***
Highest level of Education: HSc		0.15 (0.34)		
Has a bachelor degree			-1.51***	-0.71 (-0.92)
Is self-employed		-0.54 (-1.07)		
Personal income less than 50K BDT				-0.28 (-0.41)

Household characteristics

Household size		-1.73***	-1.32***	
Tenure: rented		1.2681*	1.01 (1.42)	
Residence time less than 1 year		-1.37 (-1.38)		
No. of HH members with license		-0.50 (-1.45)		
No. of employed persons in HH		1.39***	1.40***	
No. of educated persons in HH			1.43***	
HH with more than one minor			0.97*	
HH with more only one minor				0.54 (0.76)
HH with more than one senior				3.34**
HH rent less than 30K BDT		1.36 (1.53)	1.22 (1.56)	
Floor size less than 1300 sqft		-1.95***	-1.57**	
Floor size around 1300-1700 sqft				2.42***
Dwelling: separate house				0.46 (0.76)
Family income greater than 150K BDT				1.6649**
<i>Built environment / location</i>				
Log of Population density		0.39**		
Willingness to use MRT (≤500 m)		-1.18***		1.45**
Willingness to use MRT (1000-1200 m)			-0.7260*	0.76 (1.17)
Willingness to use MRT (1200-2500 m)				-2.11***
Expected Maximum Utility (EMU) of mode choice	<i>ref.</i>	0.86 (1.19)	0.61 (1.05)	0.65 (0.93)

Logarithm of commute distance to Primary Activity Location

Variable	Non-Family (reference)	Family, No Child	Family, With Child	Joint Family
Constant	<i>ref.</i>	0.56 (1.36)	1.09***	1.22***
<i>Individual characteristics</i>				
Log of age		0.25**		
Gender: male			0.03 (0.42)	
Marital status: married			-0.16**	
Education: Bachelor (BSc)			0.17***	
Is self-employed		-0.08 (-0.69)		-0.14 (-0.94)
Has license		0.23**	-0.06 (-0.86)	0.32**
<i>Household characteristics</i>				
Household size			0.05 (1.37)	
Educated person in HH (at least 1)			-0.12*	
Residence time greater than 3 years		-0.11*		
HH with minors			0.08 (0.42)	0.08 (1.01)

HH with elderly (>59)			-0.01 (-0.13)
Family income 75-150K BDT		-0.09 (-1.32)	
Family income up to 100K BDT			0.22 (1.78)
Dwelling: flat in multi-storied building			0.05 (0.51)
HH rent around 10-50K BDT		0.26***	0.08 (0.82)
<i>Expected Maximum Utility (EMU) of mode choice interacted with the following location, transit-willingness, and mobility attributes</i>			
High population density	-0.1473***	-0.1246***	-0.1025*
Residential land use	-0.1289***	-0.1633***	-0.1171*
Proximity to MRT	-0.1504***	-0.1479***	-0.1447**
HH with one or more licenses		-0.05 (-1.13)	
cycle access	-0.18 (-1.27)		
bikes in HH		-0.1386*	
private vehicles in HH			0.26 (0.71)
Number of minors and private vehicle in HH			-0.1115*
<i>Scale and dependence parameters</i>			
Std. dev. of distance, $\ln \sigma$		-1.2652***	
Correlation coefficient, ρ		0.2899*	

Notes: *t*-statistics are reported in brackets. *, **, and *** denote significance at 10%, 5%, and 1% levels, respectively; values with $|t| < 1.65$ have no stars and the *t*-value is given in the bracket. "ref." denotes the reference (base) alternative, whose utility/parameters are normalized to zero. EMU = expected maximum utility (logsum) from the mode-choice model. $\sigma = \exp(\ln \sigma) = 0.2823$; ρ is the copula correlation between the unobserved factors of living arrangement and commute distance.

4.2.2 Mode Choice Component and the Value of Travel Time Savings

The mode choice component covers the full spectrum of motorized, paratransit, and non-motorized alternatives available in Dhaka: bus, MRT, CNG auto-rickshaw, office vehicle, ride-share motorcycle, ride-share car, rickshaw, Lagoon (human hauler), bicycle, private motorcycle, private car, and walking, with bicycle serving as the reference alternative. The systematic utility of each alternative consists of an alternative-specific constant, travel duration, and travel fare or operating cost, where the travel duration and cost skims were derived from the Google Routes API for each respondent's residence–PAL pair, as described in CHAPTER 3. The purpose of this deliberately parsimonious specification is twofold: it provides a behaviorally credible representation of the relative performance of all available modes, and it ensures that the logsum derived from it is a clean, utility-consistent accessibility index that can enter the living arrangement and distance components without contamination from socio-demographic effects that already appear in those components directly (Akbari & Habib, 2015; Geurs & van Wee, 2004; Handy & Niemeier, 1997).

Both level-of-service parameters are highly significant with the theoretically expected negative signs: utility falls as travel duration increases and as fare increases. The ratio of these two coefficients implies a value of travel time savings (VTTS) of approximately 200 BDT per hour for regular commuting trips in Dhaka. This figure is plausible against prevailing urban wage levels in Bangladesh and is of the same order of magnitude as average hourly earnings implied by national per-capita income statistics, supporting the behavioral soundness of the component. It should be remembered that this is a conservative estimate referring to typical, repeated commuting conditions rather than to occasional or time-pressured trips, and that living arrangement and residential location are longer-horizon decisions for which such a representative commuting context is the appropriate basis.

The pattern of alternative-specific constants is informative about the structure of Dhaka's transport market. Holding time and cost constant, ride-share car and CNG auto-rickshaw carry the largest constants, followed by private car, rickshaw, private motorcycle, MRT, and bus, all significant at the 95 percent level, while office vehicle is also positive and significant. The high constants of the door-to-door paratransit modes reflect their unmeasured advantages in a city where walking environments are poor, transfers are onerous, and reliability of scheduled services is limited: comfort, weather protection, and the avoidance of crowding are capitalized into these constants rather than into the time and cost variables. The statistically insignificant constants of ride-share motorcycle, Lagoon, and walking indicate that, once their time and cost profiles are accounted for, these modes are perceived roughly on par with the bicycle reference. The positive and significant MRT constant, estimated only shortly after the opening of Dhaka's first metro line, suggests an unmeasured preference for the new rail mode beyond its time-cost performance, consistent with the strong stated willingness to use MRT documented in the descriptive analysis of CHAPTER 3.

From this component, the expected maximum utility (EMU) of mode choice is computed for every respondent as the logsum over all twelve alternatives. The EMU is a theoretically consistent measure of modal accessibility. It increases with the number and quality of feasible modes connecting the residence to the PAL and decreases with the time and cost of reaching it (Ben-Akiva & Lerman, 1985; Geurs & van Wee, 2004). It is this quantity, rather than any single physical proximity measure, that links day-to-day

mobility conditions to the longer-term residential decisions modeled in the remaining two components.

4.2.3 Dependence Between Living Arrangement and Commute Distance

A central feature of the model is the Gaussian copula linking the living arrangement choice and the lognormal commute distance equation. The estimated correlation coefficient ρ is 0.2899 with a t-statistic of 1.601, positive and approaching but not reaching significance at the 95 percent level. The positive sign indicates that the unobserved factors that raise the propensity of choosing a family-oriented living arrangement also tend to lengthen the commute distance to the PAL, after controlling for the rich set of observed socio-demographic, household, housing, and accessibility variables in both equations.

This result stands in instructive contrast to the strongly negative correlation (-0.93) reported for postsecondary students in Toronto, where the unobserved inclination to leave the parental home was tightly coupled with the desire to live close to campus (Hasnine et al., 2023). In Dhaka, the latent forces operate in the opposite direction. Household formation around family obligations, preferences for particular neighborhoods, school quality considerations. Above all, affordability constraints in the central housing market push family-oriented households toward locations that are farther from the primary earner's activity location. Whereas the implicit trade-off in the Toronto student market is between independence and proximity, the implicit trade-off in Dhaka is between family-centered housing consumption and proximity. The systematic variables in the model capture much of this trade-off explicitly, which is why retaining the copula link is essential, because ignoring it would force these correlated unobservables into the error terms of two separately estimated models and bias the accessibility and affordability effects of central interest (Hawkins et al., 2021; Zhang et al., 2019).

The scale parameter of the lognormal distance distribution is estimated very precisely, with $\ln \sigma = -1.2652$ ($t = -30.79$), implying $\sigma = 0.2823$. The small conditional standard deviation indicates that, once living arrangement segment membership and the systematic covariates are accounted for, commute distances are tightly distributed around their conditional means. This supports the segmentation strategy of estimating a separate distance equation for each family-oriented living arrangement: a substantial share of the raw dispersion in commute distance in Dhaka is structural, arising from who people live with, rather than idiosyncratic.

4.2.4 Living Arrangement Choice Component

The living arrangement component models four mutually exclusive alternatives: non-family households (persons living alone or with non-family members, which serves as the reference alternative), nuclear families without children under 18, nuclear families with children under 18, and joint families. The merging of the originally separate “living alone” and “with non-family members” categories into a single non-family reference was adopted after early estimation rounds revealed that the two cells were individually too small (jointly about 10 percent of the sample) to support stable alternative-specific constants. The merged category is also behaviorally coherent, as both represent residential independence from the family unit. The discussion below proceeds through the blocks of Table 2: individual characteristics, household characteristics, housing attributes, and built environment and accessibility.

4.2.4.1 Individual characteristics

The logarithm of age has a strong positive influence on all family-oriented alternatives relative to non-family living, and is significant for nuclear families without children and joint families. Life-cycle position is thus the primary individual-level sorting mechanism: residential independence in Dhaka is overwhelmingly a phenomenon of early adulthood, and the probability of being embedded in a family household, particularly a multigenerational joint family, rises steeply with age. This mirrors the life-course perspective of the residential mobility literature, in which living arrangement transitions are triggered by milestone events such as marriage and childbearing (Mulder & Clark, 2002; Rashidi & Mohammadian, 2015) but with an important contextual difference: whereas in North American settings aging is associated with leaving the family home, in Dhaka aging is associated with deeper integration into family structures.

The marital status variable reinforces this reading in a distinctly South Asian way. Being single significantly increases the probability of living in a nuclear family without children and in a joint family relative to non-family living. Unmarried adults in Dhaka, unlike their counterparts in the western student market who leave the parental home in large numbers (Hasnine et al., 2023), overwhelmingly remain within the parental or extended household until marriage. Independent living before marriage is culturally uncommon and economically difficult in Dhaka's rental market, where landlords frequently prefer family tenants over “bachelor” tenants and where mess or hostel accommodation, captured in the non-family reference, is the principal alternative for unmarried migrants.

Holding a bachelor's degree, in contrast, significantly reduces the probability of living in a nuclear family with children. Higher-educated respondents marry and bear children later, and a larger share of them are at career stages where they either remain in their family of origin or live independently near employment; the negative effect is therefore best interpreted as a life-cycle timing effect rather than as a preference against children.

4.2.4.2 Household characteristics

Household size carries significant negative parameters in both nuclear alternatives. Because the variable is omitted from the joint family utility, the negative signs indicate that, as household size grows, the arrangement is increasingly unlikely to be a compact nuclear family and correspondingly more likely to be either a large shared non-family dwelling (mess households can be large) or, especially, a joint family. The number of employed persons in the household, by contrast, significantly increases the probability of both nuclear arrangements. Multiple earners are precisely what allows a household to sustain an independent nuclear unit in Dhaka's expensive rental market, echoing the affordability findings of the Bangladeshi housing literature (Giti, 2018; Jahan & Kalam, 2012; Kamruzzaman & Ogura, 2008). The number of educated household members significantly raises the probability of nuclear families with children, reflecting the association between human capital, dual-earner capability, and independent family formation (Sultana, 2006).

The most striking household-composition effect concerns seniors. The presence of more than one senior member raises the utility of the joint family alternative very strongly. Multigenerational co-residence in Dhaka is the dominant institutional arrangement for elderly care in the near absence of formal old-age support systems, and households containing several seniors are, almost by definition, organized as joint families. This parallels, in an amplified form, the Toronto finding that the presence of members aged 75 and above keeps students in the parental home (Hasnine et al., 2023) and underscores that caregiving obligations are a first-order determinant of living arrangement in the Dhaka context. The presence of more than one minor moderately raises the probability of families with children, as expected mechanically and behaviorally.

4.2.4.3 Housing attributes

Housing consumption variables sort households across arrangements in an economically intuitive way. Small dwellings with floor area below 1,300 square feet are significantly less likely to house nuclear families of either type, while the larger floor-size category

significantly raises the probability of joint families. Space is the binding constraint of multigenerational living. Joint families require, and small units preclude, the dwelling sizes needed to accommodate extended kin, whereas the smallest units in the market are disproportionately occupied by the mess, hostel, and single-person dwellings of the non-family reference. Rented tenure raises the probability of both nuclear alternatives, consistent with the descriptive finding of CHAPTER 3 that 73 percent of sampled households rent and that independent nuclear formation in Dhaka typically occurs through the rental market rather than through ownership (Haque et al., 2020). A monthly rent below 30,000 BDT is positively associated with both nuclear forms, locating these arrangements in the mid-market rental segment, while family income above 150,000 BDT raises the probability of joint family living. High pooled incomes are partly a consequence of joint living, since multiple earning generations share one budget, and partly an enabler of the large dwellings that joint living requires. Short residence duration of less than one year lowers the probability of nuclear families without children, suggesting that recent movers are disproportionately found in the more footloose non-family category.

4.2.4.4 Built environment and accessibility

The log of population density carries a positive and nearly significant parameter for nuclear families without children. Childless couples and small adult families gravitate toward dense, central neighborhoods where small rental units and employment access coexist, a pattern consistent with international evidence on the residential preferences of small households (Guo & Bhat, 2007; Schwanen & Mokhtarian, 2005).

The stated willingness to use MRT at alternative station distances yields one of the most policy-relevant patterns in the component. Willingness to use MRT when a station lies within 500 meters of the residence significantly lowers the probability of nuclear families without children but significantly raises the probability of joint families. Willingness conditional on stations at 1,200–2,500 meters significantly lowers joint family probability. Read together, these parameters indicate that large multigenerational households are strongly transit-oriented but only under genuinely walkable station access. With many members, diverse destinations, and low per-capita vehicle availability, joint families value a high-capacity trunk mode at their doorstep, yet their willingness evaporates once station access requires a long walk or a feeder trip. Small adult households, by contrast, are less dependent on any single trunk mode, retaining

flexible access to the full paratransit spectrum. This differentiated transit orientation by household structure has direct implications for the land-use planning of MRT station areas, developed in Section 4.5.

Finally, the expected maximum utility of mode choice enters all three family-oriented utilities with positive parameters. Directionally, better modal accessibility between home and PAL is associated with family-oriented arrangements relative to non-family living: when the transport system performs well, workers do not need to detach from the family household and move into proximate mess accommodation in order to manage the commute. The modest significance indicates that, in Dhaka, accessibility shapes living arrangement primarily indirectly, through the distance margin examined next, rather than through the discrete arrangement margin itself. This is a meaningful contrast with the western living style, where the EMU was a strong direct determinant of leaving the parental home (Hasnine et al., 2023) and reflects the far stronger cultural anchoring of household structure in the Dhaka context.

4.2.5 Commute Distance Component

The commute distance component models the natural logarithm of the distance between the residence and the PAL as a lognormal regression with a separate equation for each of the three family-oriented living arrangement segments; the non-family reference carries no distance equation, and its members' distances are accommodated through the joint likelihood structure. The lognormal assumption respects the non-negativity and right skew of observed commute distances (mean 7.6 km, standard deviation 5.8 km in the estimation sample). Within each segment, the equation contains individual and household covariates together with a set of interactions between the EMU and indicators for the respondent's stated primary reason for choosing the current residence. The interaction structure deserves emphasis, because the EMU does not enter the distance equations as a standalone main effect, but only as moderated by the dominant residential motivation. So, the accessibility sensitivity of commute distance is allowed to differ according to why the household located where it did.

4.2.5.1 Constants and individual characteristics

The segment constants are positive and significant for families with children and joint families, and positive but insignificant for families without children, establishing that family-oriented households, particularly those with children and extended kin, systematically maintain longer commutes than the covariate-adjusted baseline. The log

of age significantly lengthens commute distance for families without children. Older members of childless family households are typically established earners with longer job tenure who have accumulated housing arrangements that they do not readjust as employment locations shift, a pattern consistent with the positive age–distance gradients found in previous literature (Hasnine et al., 2023; Zhang et al., 2019) and with commuting-tolerance findings in the excess-commuting literature (Giuliano & Small, 1993). Married respondents in families with children commute significantly shorter distances. Once children anchor the household to schools and child-related amenities, the married principal earner economizes on commute length, whereas unmarried adult members of such households (typically older siblings or relatives) absorb longer commutes because the residence is optimized around the children's rather than their own activity locations. Holding a bachelor's degree significantly lengthens the commute of respondents in families with children. Higher-educated workers access spatially concentrated formal employment, especially in Dhaka's central business and administrative districts, and can monetarily sustain the associated commute, consistent with the higher commute tolerance of high-skill workers reported internationally (Masoumi, 2021; Yao & Wang, 2014).

Possession of a driving license is among the most consequential individual covariates, significantly lengthening commute distance in both the family without children segment and the joint family segment. A driving license indicates access to private motorized mobility, such as cars, motorcycles, or ride-sourcing, which reduces the spatial limitations created by Dhaka's slow surface transport. Licensed individuals can convert speed into space, choosing more distant residences with better housing value. The same mechanism has been documented repeatedly in the residential location literature, where car availability decouples residence from workplace (Pinjari et al., 2011; Schwanen & Mokhtarian, 2005). Notably, the corresponding parameter for families with children is small and negative, suggesting that the school anchor neutralizes the distance-enabling effect of motorization in child-rearing households.

4.2.5.2 Household and housing covariates

Household size modestly lengthens commutes in the family with children and is essentially flat for joint families, while the presence of at least one additional educated member shortens commutes in the family with children segment. Multi-earner, multi-educated households face multiple workplace anchors and resolve them by centralizing

the residence, a finding that resonates with the jobs–housing balance literature on dual-earner households (Abraham & Hunt, 1997; Akbari & Habib, 2015; Sultana, 2006). Residence duration above three years shortens commutes for families without children, consistent with gradual spatial adjustment toward employment over time. Among the housing variables, the most important effect is monthly rent in the 10,000–50,000 BDT band, which significantly lengthens commutes in the family with children segment and positively, though insignificantly, in the joint family segment. This is the clearest direct expression of the housing-affordability–commute trade-off in the model. Families with children seeking dwellings in the dominant mid-rent band are priced out of locations near the major employment and education concentrations and accept measurably longer commutes in exchange for attainable rents, exactly the location-affordability mechanism emphasized in the housing–transportation cost literature (Dewita et al., 2020; Howell et al., 2018; Makarewicz et al., 2020; Stone & Stone, 2010) and previously identified for Dhaka commuters by (Choudhury & Ayaz, 2015a). Family income up to 100,000 BDT lengthens joint family commute, indicating that it is the moderate-income joint families, rather than the most affluent ones, that are pushed to peripheral locations.

4.2.5.3 Accessibility interactions

The block of EMU interaction terms constitutes the analytical core of the distance component and one of the principal contributions of this thesis, so its construction deserves careful description before its estimates are read. The expected maximum utility does not enter the distance equations on its own, rather it enters multiplied by a set of indicators that describe, for each household, the measured context in which that accessibility is experienced. The indicators come from three distinct sources, and none of them is a stated “reason” for choosing the residence. The first source is objective spatial measurement. This describes where the household factually lives, independently of anything the respondent believes or reports about why. The second source is a stated-response scenario question on the new metro system. The MRT willingness indicator measures the household's behavioral orientation toward rail transit, not its residential motivation. The third source is the household's own mobility endowment: driving licenses, bicycle access, motorcycles, and private vehicles. Because the indicators originate from different instruments, they are not mutually exclusive. A single household can simultaneously live in a dense residential area, declare itself willing to use the MRT, and hold a license, in which case its total accessibility sensitivity is the sum of the

applicable interaction terms.

This construction has two methodological consequences worth stating explicitly. First, it allows the marginal effect of accessibility on commute distance to vary across residential contexts instead of imposing a single citywide elasticity. Accessibility does not mean the same thing in a dense, mixed inner thana, where high EMU reflects a genuine abundance of nearby modes and destinations, as it does on the low-density periphery, where a similar logsum may be produced by a few long but fast connections. Interacting the EMU with measured density and land use lets the data assign different distance responses to these different situations (Geurs & van Wee, 2004; Handy & Niemeier, 1997). Second, the interaction form disciplines a well-known confounding risk. The EMU is built from travel times and costs that are themselves correlated with density, so entering accessibility and density as separate main effects would invite collinearity and ambiguous attribution. By letting density and land use condition the slope of the accessibility effect rather than compete with its level, the specification extracts the behavioral content of accessibility within each context. The price of this design is a deliberate restriction: a household exhibiting none of the interacted attributes carries, by construction, a zero accessibility effect in its distance equation, a benchmark against which the estimated interactions should be read.

With the construction clear, the estimates can be given their proper weight. All three context indicators produce consistently negative interactions across the family segments. For households residing in high population density areas, greater modal accessibility significantly shortens the commute among nuclear families without children and with children, with a negative but insignificant effect for joint families. The behavioral reading is that density and accessibility are complements in compressing urban space where the residence sits amid a thick fabric of opportunities, every improvement in the performance of the modes serving it translates into actually living closer to the Primary Activity Location, because proximate alternatives exist to be chosen. The same accessibility improvement on the thin periphery has nothing nearby to pull the household toward. The negative and significant sign of this interaction, recovered here in dummy form for numerical stability, is theoretically consistent with the density–accessibility complementarity emphasized in the built-environment literature (Bhat & Guo, 2007; Guan & Peiser, 2018).

For households whose geocoded residence falls in a predominantly residential land-use

zone, the accessibility interactions are likewise negative and significant for the two nuclear segments and negative for joint families. It is notable that the strongest of all the land-use effects belongs to families with children in residential zones. These are precisely the households for whom an orderly residential environment, with its schools, playgrounds, and quieter streets, bundles daily destinations close to home, so that good modal accessibility allows the entire pattern of household life, including the respondent's own commute, to be internalized within a compact radius. The result gives empirical content to the planning intuition that residential zoning succeeds, in travel terms, only in combination with transport supply. The zone retains its residents' travel locally when, and to the degree that, it is accessible (van Heerden et al., 2022; Waldeck et al., 2020).

The most striking row of the block, however, is the MRT willingness interaction. For respondents who state that they would use the metro for their daily trips were a station within walking distance, greater modal accessibility significantly shortens the commute in *all three* family segments and the three coefficients are remarkably uniform, spanning barely 0.005 in magnitude. Two features of this result deserve emphasis. First, it is the only accessibility channel that remains significant for joint families, the household type whose residence, as Sections 4.3.4 and 4.4.3 establish, is otherwise the most immovably fixed by collective considerations. Even for them, a declared orientation toward rail transit is the condition under which accessibility gains convert into proximity. Second, the near-identity of the three coefficients indicates that transit orientation operates as a universal accessibility channel, indifferent to household structure, in sharp contrast to the density and land-use channels, whose strength varies across segments. Where the Toronto study found walking and cycling accessibility to be the dominant compressing channel for students (Hasnine et al., 2023), in Dhaka the role passes to rail willingness, exactly as should be expected in a megacity where the pedestrian environment is hostile, bicycles are present in only 3 percent of households, and the new MRT line constitutes the first reliable escape from congested surface travel. To give the magnitudes a physical sense: a coefficient near -0.15 implies that a one-point improvement in the logsum reduces the expected commute of a transit-willing household by roughly 14 percent, about one kilometer at the sample mean. For a household that additionally resides in a high-density residential zone, the applicable interactions accumulate to approximately -0.43 in the family-with-children segment, so the same accessibility improvement compresses its expected commute by more than a third. The negative significant

interactions also carry the converse implication: transit-willing and density-situated households whose locations are poorly served commute systematically farther, quantifying the distance penalty currently borne by precisely the households most ready to reward transit investment with shorter journeys.

The mobility-endowment interactions complete the block and are individually weaker but jointly informative. The interactions of the EMU with household licenses, bicycle access, and motorcycles in the household are all negative. Even among households holding private mobility options, improvements in overall modal accessibility are associated with shorter, not longer, commutes, indicating that accessibility gains are not simply absorbed as license-enabled sprawl. The single positive interaction, private vehicles in joint families (0.262, $t = 0.72$), is statistically indistinguishable from zero, while the interaction of minors together with private vehicles in joint families is negative (-0.112 , $t = -1.63$), hinting that vehicle-owning multigenerational households with children deploy accessibility gains to shorten the household's journeys rather than to extend its reach. Read against the strong standalone license effects of the distance equations (Section 4.2.5) and of the marginal effects to follow (Section 4.4), the pattern resolves a potential paradox: motorization lengthens commutes through its direct, residence-enabling channel, but conditional on where the household lives, better accessibility moderates rather than amplifies that lengthening. Finally, the additive structure of the non-exclusive indicators should be borne in mind when reading these estimates jointly: the model represents each household's accessibility responsiveness as the sum of the channels that apply to it, a flexible representation whose robustness to alternative groupings of the indicators is a natural subject for the sensitivity analyses proposed in CHAPTER 5.

4.3 Marginal Effects of Living Arrangement Model

The parameter estimates in Table 4.1 tell us the direction and the statistical strength of each effect, but they do not yet tell us how much each factor actually matters for the choices people make. In a multinomial logit model, the effect of any variable on the probability of an alternative depends on the entire set of utilities and on each person's baseline probabilities, so a coefficient of, say, 2.4 cannot be read directly as a probability change. To bridge this gap, probability-weighted sample enumeration (PWSE) marginal effects were computed for every binary explanatory variable in the living arrangement

component. The procedure is intuitive: for each respondent in the sample, the variable of interest is switched from 0 to 1 while everything else about that person is left exactly as observed. The model is then asked how the predicted probabilities of the four living arrangements change and finally these changes are averaged over the whole sample, weighting each respondent by their predicted choice probabilities. The resulting numbers can be read in plain language as they are the average change, in percentage points, in the probability of each living arrangement when a person acquires the attribute in question. Because probability gained by some alternatives must be lost by others, the effects of each variable sum approximately to zero across the four alternatives, which serves as a built-in sanity check on the computation. This discrete-change approach is the analogue of an elasticity for dummy variables, for which infinitesimal elasticities are not defined.

Figure 4.1 through Figure 4.4 visualize these effects for the four alternatives in turn. Rather than walking through the variables in the order they appear in the table, the discussion below reads each figure as a portrait. Each living arrangement in Dhaka has a recognizable face, a typical dwelling, a typical rent, and a typical relationship to the transport system, and the marginal effects allow these portraits to be drawn with numbers attached. Reading the four portraits side by side then reveals how the same variable, a second senior in the household, a small flat, an MRT station around the corner, pushes people out of one arrangement and into another, which is precisely the kind of substitution that planners need to understand.

4.3.1 Non-Family Households

The non-family alternative collects everyone who lives outside a family unit: people living alone, and people sharing mess, hostel, or other non-family accommodation. In the sample this is a small world, about one respondent in ten, but it is a distinctive one, and the marginal effects in Figure 4.1 sketch its boundaries with remarkable clarity.

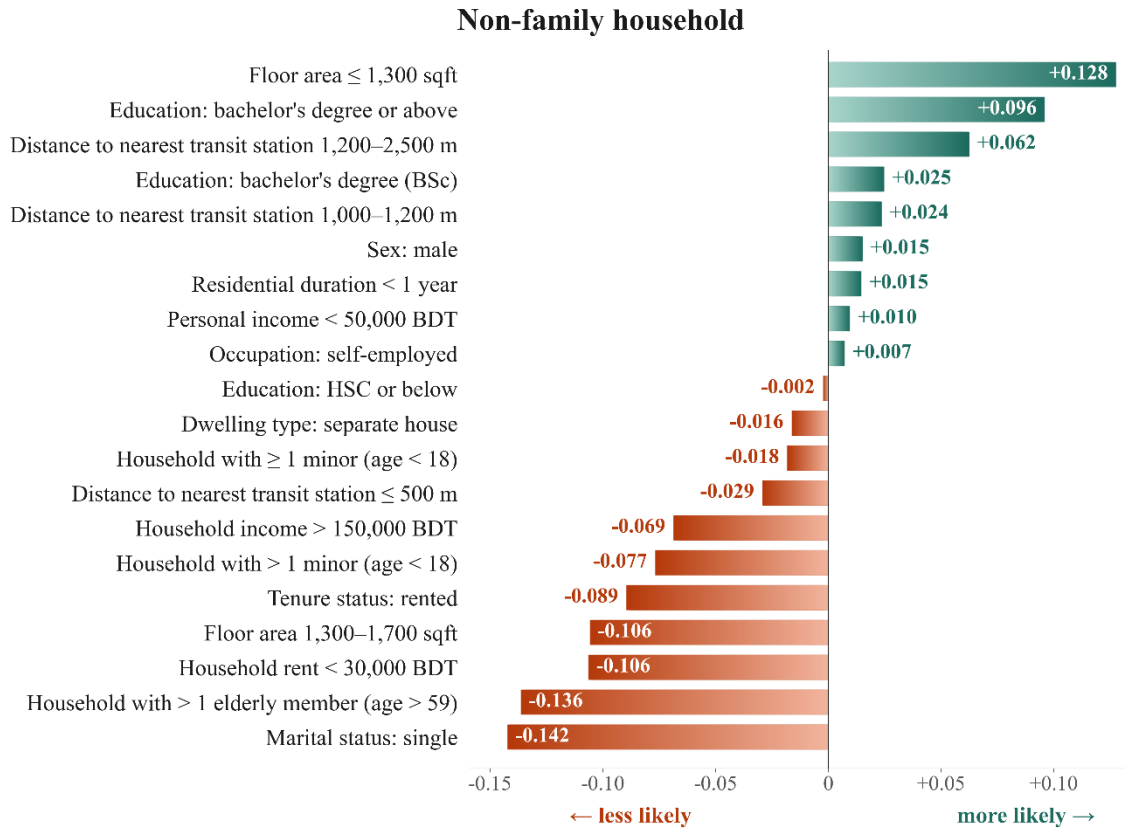


Figure 4.1: Marginal Effects of Socio-Demographic and Spatial Factors on Non-Family Household Choice

The strongest force pulling people into this arrangement is the dwelling itself. Occupying a unit with floor area below 1,300 square feet raises the probability of non-family living by 12.8 percentage points, the largest positive effect for this alternative. The smallest units in Dhaka's housing stock, single rooms, subdivided flats, mess seats, are physically incapable of housing a family, and so they are occupied by those who live without one. Education tells the second part of the story: holding a bachelor's degree or higher raises the probability by 9.6 percentage points. The picture that emerges is the familiar one of the educated migrant. A graduate who has come to Dhaka for a formal-sector job, whose family remains in the district town, and who solves the housing problem with a shared flat or mess seat near work. Even the transit variable fits this portrait. Stated willingness to use MRT at the most distant access band (1,200–2,500 m) adds 6.3 percentage points to non-family probability. A single person with a flexible schedule and no school runs to manage can tolerate a long walk or a rickshaw feeder to the station in a way that a family household, as will be seen below, simply will not.

The forces pushing people out of non-family living are, at first glance, surprising, and

they are among the most culturally diagnostic results in the thesis. Being single reduces the probability of living outside a family by 14.2 percentage points. In most Western evidence, singlehood and solo living go hand in hand. Young adults leave the parental home precisely because they are unattached (Hasnine et al., 2023; Mulder & Clark, 2002). In Dhaka the relationship runs the other way. An unmarried adult is, by social default, a member of the parental or extended household. Independent living before marriage is culturally uncommon, and the rental market reinforces the norm, since landlords visibly prefer family tenants over “bachelors.” The single people who do appear in the non-family sector are there because migration forced the issue, not because singlehood invited it. Family obligations exert the same inward pull. The presence of more than one senior in the household reduces non-family probability by 13.6 percentage points, and more than one minor reduces it by 7.7 points. A person embedded in a household that contains elderly parents or several children does not peel off into a mess. The household needs them, and they stay.

The negative effects describe what the non-family sector is not. It is not the mainstream rental market. Rent below 30,000 BDT per month reduces non-family probability by 10.6 percentage points, the mid-size floor band by 10.6 points, rented family tenure by 9.0 points, and family income above 150,000 BDT by 6.9 points. Each of these attributes belongs to the world of established family households, mid-rent family flat, the pooled multi-earner budget, and acquiring any of them signals a move out of bachelor accommodation. Taken together, Figure 4.1 portrays the non-family sector as a separate housing submarket. Small units, educated single occupants, high turnover, and a tolerance for imperfect transit access. It functions as a gateway for Dhaka’s migrant workforce and, as Section 4.5 argues, warrants regularization rather than neglect.

4.3.2 Nuclear Families without Children

The second portrait belongs to the nuclear family without children under 18, which is the largest single category in the sample at 36 percent. It is important to be precise about who lives here, because the label is slightly deceptive. The category contains young childless couples, certainly. But it also contains parents whose children have all crossed the age of 18 and who therefore, by the survey definition, no longer count as minors, together with those grown children themselves. The marginal effects make this composition visible.

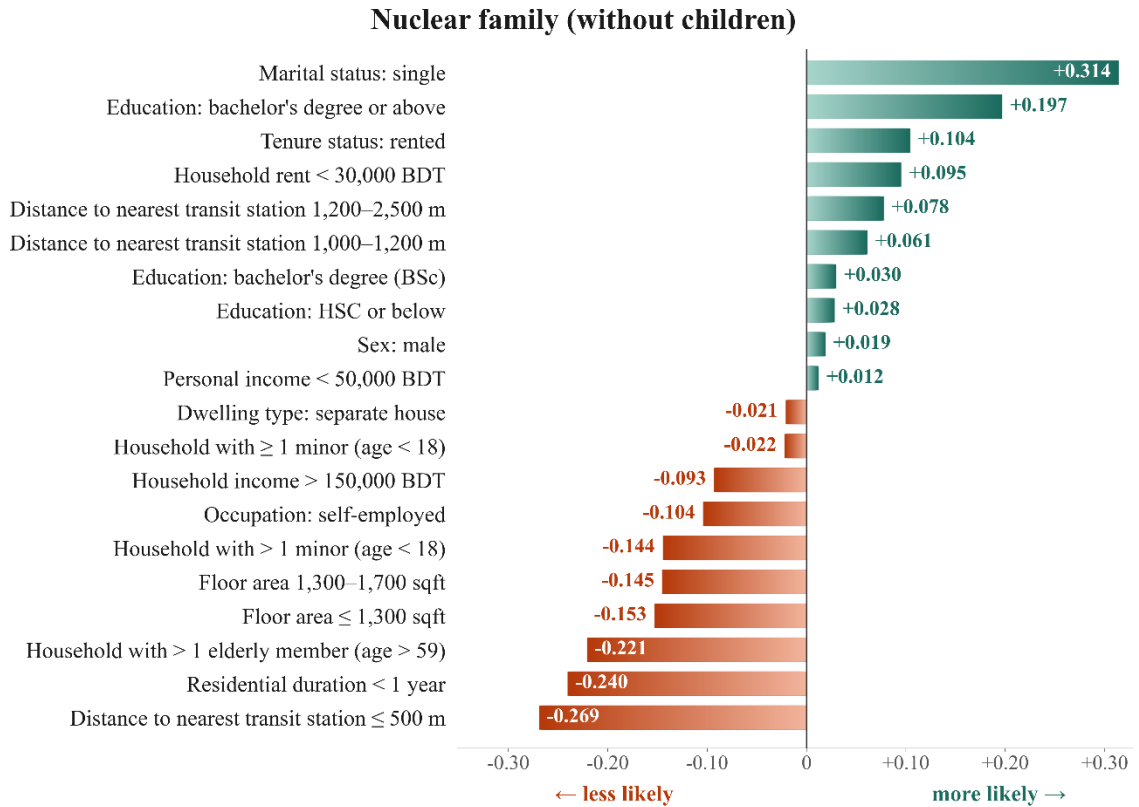


Figure 4.2: Marginal Effects of Socio-Demographic and Spatial Factors on Nuclear Family (Without Children) Choice

From Figure 4.2 we can see that being single is the strongest positive effect in the entire figure, raising the probability of this arrangement by 31.4 percentage points. These are not single people living alone. Section 4.3.1 has just shown that singles avoid solo living. They are unmarried adult sons and daughters who remain in the parental flat through their twenties and beyond, waiting for marriage, for a posting, or simply for housing they can afford. The educated adult child living with parents is one of the defining household forms of urban Bangladesh, and the model recovers it with one of its largest effects.

The housing variables place this arrangement squarely in the rental mainstream. Rented tenure raises its probability by 10.4 percentage points and rent below 30,000 BDT by 9.5 points. The minor less nuclear family is the typical occupant of the mid-rent family flat that dominates Dhaka's housing stock, the market segment whose affordability stress has been the central concern of the Bangladeshi housing literature (Jahan & Kalam, 2012; Kamruzzaman & Ogura, 2008). Both the small floor size and the larger floor size reduce its probability, locating these households in intermediate dwelling sizes, larger than a mess seat, smaller than a joint family house. One temporal effect deserves a sentence of its own: residence duration of less than one year reduces the probability of this

arrangement by 24.0 percentage points. Recent movers are concentrated in the footloose non-family sector. The minor less nuclear family, by contrast, is residentially settled, often having occupied the same flat since before the children grew up.

The most striking negative effect in Figure 4.2, however, concerns the metro. Stated willingness to use MRT when a station lies within 500 meters reduces the probability of this arrangement by 26.9 percentage points, the mirror image of the large positive effect for joint families discussed in Section 4.3.4. The interpretation is not that childless families dislike the MRT, it is that close-in trunk transit is not what defines them. A compact household of two or three adults, with flexible schedules and few simultaneous destinations, can stitch its travel together from Dhaka's full paratransit spectrum, the bus, the CNG, the rickshaw, the ride-share motorcycle, and does not need to organize its residential location around a single high-capacity mode. Demographic composition completes the picture in the expected direction: more than one senior and more than one minor shift households out of this category and into the joint and child-rearing categories respectively, which is where the narrative now turns.

4.3.3 Nuclear Families with Children

If the childless nuclear family is defined by waiting, the family with children is defined by commitment, and the marginal effects in Figure 4.3 are dominated by the two great commitments of marriage and parenthood. Being single reduces the probability of this arrangement by 47.4 percentage points. This is the largest marginal effect anywhere in the living arrangement model, and it is worth pausing on. Nearly half of the probability mass of this alternative is governed by a single life-course attribute.

In Dhaka, children arrive inside marriage, and marriage is the gateway through which households enter this category. The model is, in effect, measuring the demographic regime of the city. Education works on the same life-course clock. A bachelor's degree or higher reduces the probability by 25.0 percentage points. Educated respondents marry later, bear children later, and spend more of their observed adult years in the childless and non-family categories. The effect is a statement about timing, not about preferences against children.

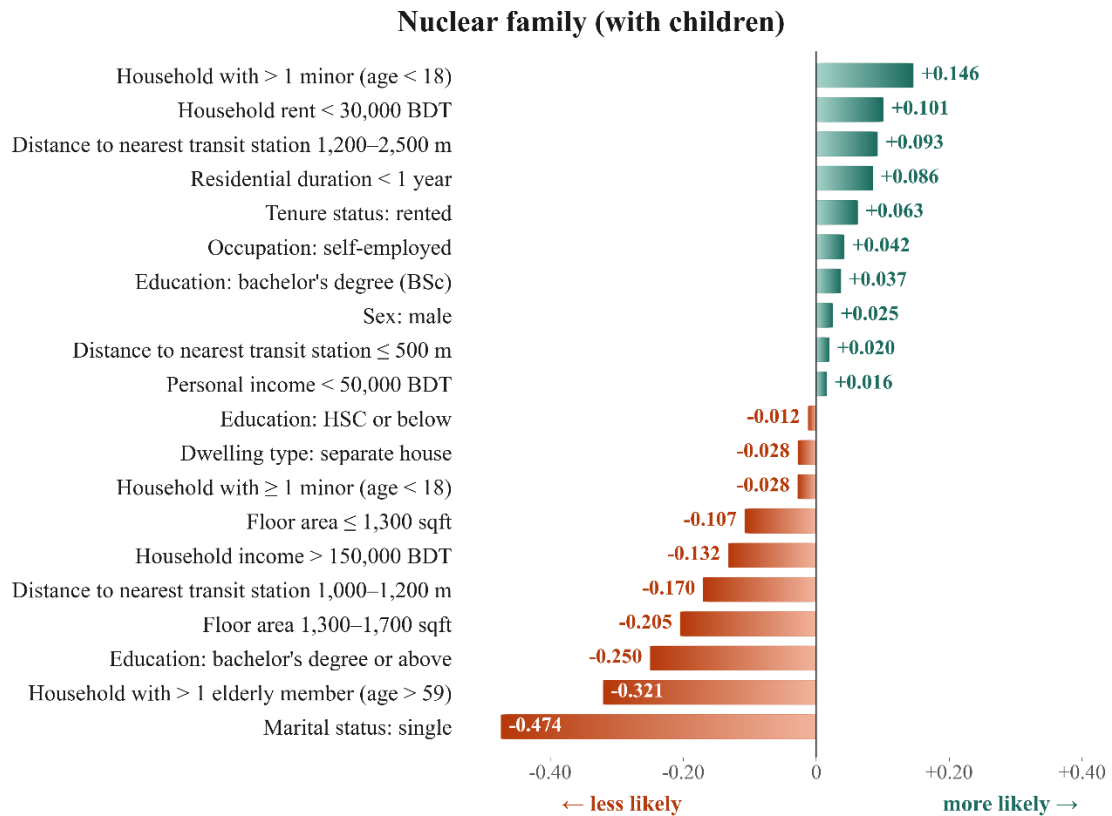


Figure 4.3 : Marginal Effects of Socio-Demographic and Spatial Factors on Nuclear Family (With Children) Choice

The third large negative effect, more than one senior in the household, is a matter of classification with real behavioral content. Once several elders live under the same roof as the children, the household is no longer a nuclear family at all but a joint one, and the probability mass migrates accordingly.

The positive effects describe the texture of child-rearing life in the city. The presence of more than one minor raises the probability by 14.6 percentage points, an effect that is partly definitional but also behaviorally informative. A household with several children is locked into the family form, unable to dissolve into other arrangements. The housing effects again point to the affordable rental mainstream, rent below 30,000 BDT adds 10.1 points and rented tenure adds 6.3 points, but with one telling addition: residence duration of less than one year *adds* 8.6 points to this arrangement, where it subtracted 24 points from the childless one. Families with children move, and they move for identifiable reasons such as a better school catchment, a bigger room as the children grow, a rent the family can still carry. The school-driven character of these moves is exactly what the school-travel literature would predict (Ewing et al., 2004) and it foreshadows the commute results of Section 4.4.2, where the children's anchors will be seen pulling the

respondent's own commute outward.

The transit variable adds a final, subtle stroke to the portrait. Willingness to use MRT at the intermediate access band of 1,000–1,200 meters reduces the probability of this arrangement by 17.0 percentage points. A kilometer's walk to the station is a different proposition with a school-age child in tow than it is for a single adult. Child-rearing families are not the marginal market for medium-walk transit access, and their travel remains organized around door-to-door modes and school proximity. Where the larger floor band reduces this arrangement's probability by 20.5 points, the household has usually crossed into joint family territory, and it is to that final empirical portrait that the discussion now turns.

4.3.4 Joint Families

The joint family is the most distinctive household form in the model and, in many ways, the most distinctively South Asian. Seventeen percent of the sample lives in one, and Figure 4.4 shows that its probability is organized around three pillars.

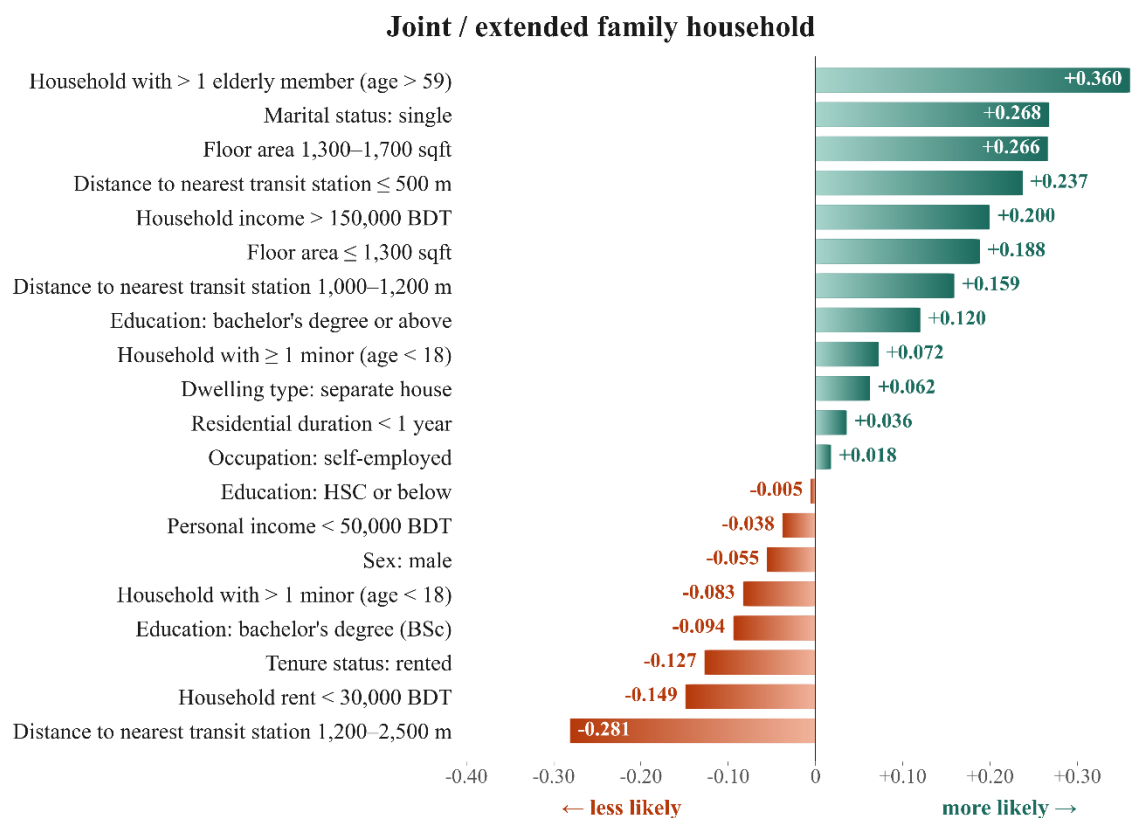


Figure 4.4: Marginal Effects of Socio-Demographic and Spatial Factors on Joint Family Choice

The care of elders, the consumption of space, and a close relationship with high-capacity

transit. The first pillar towers over the others. The presence of more than one senior in the household raises joint family probability by 36.0 percentage points, the largest positive marginal effect in the entire model. In a city with essentially no formal old-age care infrastructure, the multigenerational household is the elderly-care system, and the model quantifies that institution. When a second elder is present, more than a third of the household's probability mass shifts into the joint form. The same gravitational pull appears, at smaller magnitude, for minors. At least one minor adds 7.2 points, as grandchildren and grandparents tend to arrive in the same household. And singlehood adds 26.8 points, completing the pattern established across the previous portraits, since the unmarried adult who does not live alone and is not raising children is very often absorbed into the extended family of a married sibling or of the parents themselves.

The second pillar is space and the means to afford it. The larger floor band raises joint family probability by 26.6 percentage points, the separate house by 6.3 points, and family income above 150,000 BDT by 20.0 points. The economics of joint living run in both directions at once. Pooled incomes from multiple earning generations are what make a large dwelling attainable, and the large dwelling is what makes multigenerational co-residence physically possible. The negative housing effects are the mirror of the rental mainstream: rent below 30,000 BDT reduces joint probability by 14.9 points and rented tenure by 12.7 points, because joint families are disproportionately found in owned, often inherited, property rather than in the mid-rent rental flats of nuclear households. There is, however, a discordant note that honest reporting should not smooth over. The smallest floor band, below 1,300 square feet, also raises joint probability, by 18.8 points. Alongside the prosperous joint family in its large owned house there evidently exists a second, constrained population of extended families crowding into small units, multigenerational not by preference but by necessity, the kind of affordability-stressed doubling-up that the Dhaka housing literature has long warned about (Jahan & Kalam, 2012; Kamruzzaman & Ogura, 2008). The joint family category, in other words, contains both the city's most comfortable and some of its most crowded households.

The third pillar is the metro, and it yields the sharpest spatial threshold in the thesis. Willingness to use MRT when a station lies within 500 meters raises joint family probability by 23.7 percentage. At 1,000–1,200 meters the effect is still positive, at 15.9 points, but at 1,200–2,500 meters it reverses violently, reducing joint probability by 28.1 points. The behavioral logic is straightforward once the household is pictured concretely.

A joint family contains many travelers, school-going children, working adults of two generations, elders with medical appointments, and very few vehicles. For such a household, a high-capacity trunk mode at the doorstep is transformative, serving every member's diverse destination at once. But the value of that mode decays with every additional minute of first-mile walking, multiplied across every member and every trip, and somewhere past 1.2 kilometers it is extinguished entirely (Hasnine et al., 2019). No other household type shows this gradient, and Section 4.5.2 builds its transit-oriented development recommendations directly upon it.

Read across the four figures, the marginal effects resolve into a single coherent narrative of substitution. The second senior who adds 36 points to the joint family takes 32 of them from the family with children, 22 from the childless family, and 14 from the non-family sector. The small flat that adds 13 points to non-family living takes 15 from the childless family and 11 from the child-rearing one. The nearby MRT station that adds 24 points to the joint family takes 27 from the childless nuclear household. Living arrangements in Dhaka are not independent lifestyle choices. They are connected vessels, and the model shows precisely along which pipes the probability flows when demography, housing, or transit changes.

4.4 Marginal Effects of Travel Distance model

The distance component of the joint model answers a different kind of question. The living arrangement effects of Section 4.3 concern who lives with whom, but the distance effects concern how far from the Primary Activity Location that household ends up, and they are measured not in percentage points of probability but in kilometers of daily life. To obtain them, the same probability-weighted sample enumeration logic was applied to the distance equations. For each binary covariate, the predicted commute distance of every respondent in the relevant segment is evaluated with the covariate switched off and then on, holding everything else at observed values and applying the lognormal retransformation, and the differences are averaged with probability weights. For the count variables, household size, the number of minors, the reported effects refer to a one-unit increase, the arrival of one more person of the given kind. Two structural exclusions apply. The non-family reference category carries no distance equation, so no effects are reported for it; its members' distances are accommodated through the joint likelihood. And the EMU interaction terms are excluded from this exercise, because their effects

operate through the continuous accessibility index and are more faithfully read through the interaction parameters discussed in Section 4.2.5.

A useful yardstick for everything that follows is the sample mean commute of 7.6 kilometers. An effect of one kilometer is therefore not a rounding error. It is roughly an eighth of the average Dhaka commute, repeated twice a day, six days a week, on streets where the mode choice model has just priced travel time at about 200 BDT per hour. Figure 4.5 through Figure 4.7 present the effects for the three family-oriented segments, and the discussion reads each figure, as before.

4.4.1 Nuclear Families without Children

The childless nuclear family has the simplest spatial problem of the three segments, no school runs, no elders' clinics, and Figure 4.5 shows that its commute geography is governed by a correspondingly simple currency which are motorized mobility.

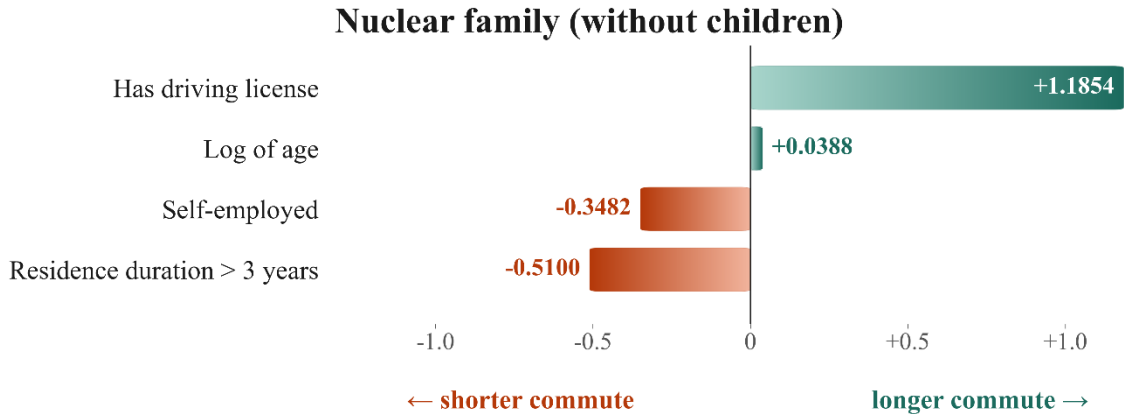


Figure 4.5: Marginal Effects of Socio-Demographic Factors on Commute Distance for Nuclear Families (Without Children)

The dominant effect, by a wide margin, is the driving license, which lengthens the average commute by 1.19 kilometers, roughly 16 percent of the mean. The mechanism is the oldest one in urban economics. A license stands for effective access to individual motorized travel, a household motorcycle, a car, a ride-share account used without hesitation, and that access converts speed into space: the licensed household can contemplate a flat in a quieter, cheaper, farther neighborhood that an unlicensed household, condemned to the bus queue and the rickshaw, cannot.

The compressing forces in this segment are forces of settledness. Residence duration above three years shortens the commute by 0.51 kilometers. Households that have stayed put have either chosen well in the first place or have watched employment, markets, and

services accrete around them, the gradual spatial adjustment that the excess-commuting literature describes as the slow grinding down of mismatch over time (Giuliano & Small, 1993). Self-employment shortens the commute by 0.35 kilometers, and the reason is visible on any Dhaka Street. The self-employed run shops below their flats, workshops in the next lane, businesses woven into their own neighborhoods. They possess an adjustment margin, moving the work to the home, that no salaried employee of a bank or a garment office can exercise. The age effect, finally, is small but positive (about 0.04 kilometers per log-unit of age), the faint trace of older earners whose long-settled housing no longer tracks their employment, consistent with the significant positive age parameter discussed in Section 4.2.5. The portrait, in sum, is of a footloose household whose commute is a chosen quantity. Lengthened deliberately in exchange for space when motorization permits, and shortened gradually by tenure and by the home-anchored economy of self-employment.

4.4.2 Nuclear Families with Children

The arrival of children transforms the household's spatial problem, and Figure 4.6 is correspondingly the richest of the three distance portraits. Its largest effect is not motorization but money, or more precisely the lack of it in the right places.

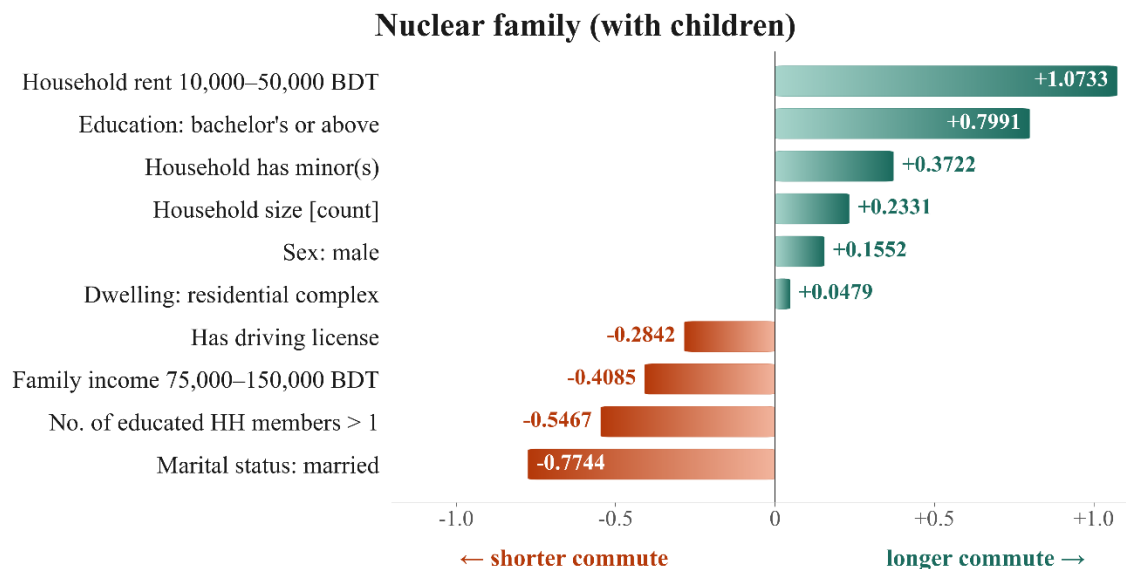


Figure 4.6: Marginal Effects of Socio-Demographic Factors on Commute Distance for Nuclear Families (With Children)

Monthly rent in the 10,000–50,000 BDT band, the band in which 72 percent of all sampled households live, lengthens the commute of families with children by 1.07 kilometers, about 14 percent of the mean commute. This is the location-affordability

trade-off of the housing literature rendered in its most concrete form (Dewita et al., 2020; Makarewicz et al., 2020; Stone & Stone, 2010). The family that needs an affordable two-bedroom flat near the schools and the offices cannot find one there, because land near Dhaka's activity concentrations is priced beyond the mid-rent band, and so the family takes the same flat four stops farther out and pays the difference every morning in travel. Choudhury & Ayaz (2015) asked of Dhaka's commuters, "why live far?"; for the child-rearing nuclear family, this single number is a large part of the answer. The effect is reinforced from the human-capital side. A bachelor's degree adds 0.80 kms, because educated parents commute to the spatially concentrated office districts that anchor the city's formal economy, and they possess the earning power to sustain the journey (Masoumi, 2021). The presence of a minor adds 0.37 kilometers and each additional household member adds 0.23 kilometers. The residence is being optimized around the children's schools and the household's collective needs, and it is the respondent's own commute that absorbs the residual.

What makes this segment behaviorally fascinating, however, is the strength and the logic of its compressing forces, several of which reverse the signs found in the other segments. Marriage shortens the commute by 0.77 kilometers, the counterpart of the significant negative parameter of Section 4.2.5, and the contrast it implies is sharp. Within child-rearing households, the married principal earners centralize the household around their joint constraints, while the unmarried adult members, the elder son working across town, the sister teaching in another thana, accept the longest journeys, because the residence was never chosen for them. A second educated member shortens the commute by 0.55 kilometers. Two graduate earners mean two workplace anchors, and the household resolves the tension by centralizing, the dual-earner balancing act documented in the jobs–housing literature (Abraham & Hunt, 1997; Akbari & Habib, 2015; Sultana, 2006). Family income of 75,000–150,000 BDT shortens it by 0.41 kilometers, comfortable households simply buying proximity.

Most telling of all is the license. In the childless segment a license lengthened the commute by 1.19 kilometers, but here it shortens it by 0.28. The instrument is the same, motorized flexibility, but the household has changed, and with it the use to which flexibility is put. The school anchor fixes the residence. The licensed parent uses the vehicle not to move farther away but to knit together the school drop-off, the office, and the market within a compact radius. Motorization, the model is saying, does not have a

sign of its own. Its sign is given to it by the household structure within which it is exercised. One minor effect completes the portrait. Being male adds 0.16 kilometers, the modest gender gap in commute length that household-responsibility patterns produce in developing-city contexts (Venter et al., 2007).

4.4.3 Joint Families

The joint family's residence is the most over-determined location in the city. It must house two or three generations. It must sit near the grandchildren's schools and within reach of the elders' familiar surroundings. It is frequently owned rather than rented, and therefore nearly immovable and it must do all of this with the lowest per-capita vehicle availability of any household type.

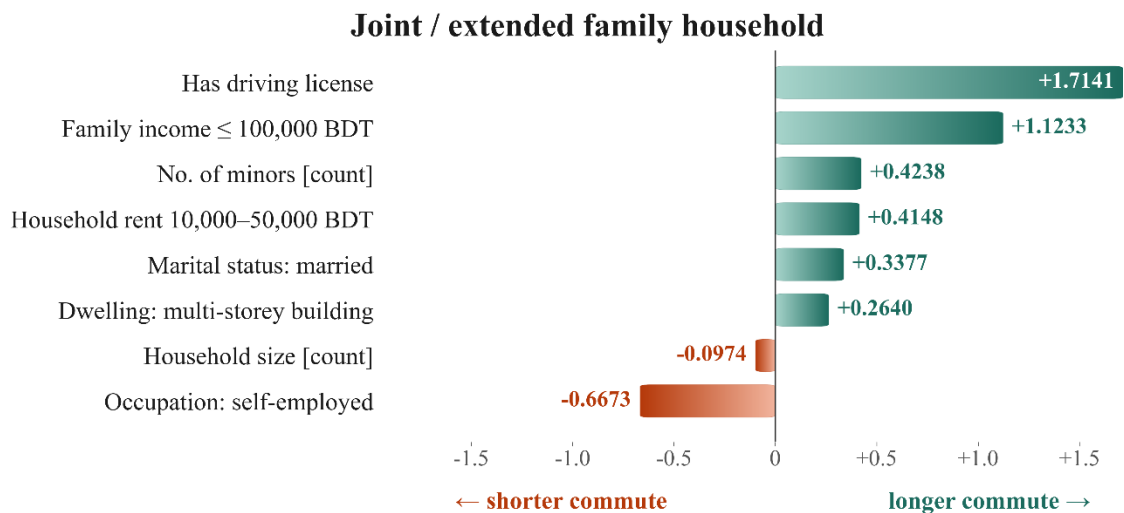


Figure 4.7: Marginal Effects of Socio-Demographic Factors on Commute Distance for Joint Families

The one thing the joint family's location is not chosen for is the individual respondent's commute, and Figure 4.7 shows the consequences. This segment exhibits the largest distance effects in the entire model. The driving license leads, lengthening the commute by 1.71 kilometers, almost a quarter of the mean commute and the single largest marginal effect in the distance model. The interpretation follows directly from the household's logic. When a residence is fixed by collective considerations, someone must bridge the gap between that fixed point and the dispersed workplaces of the household's earners, and the model identifies who. The licensed, motorized member, typically a mid-generation working adult, who absorbs the household's spatial mismatch on a motorcycle or behind a wheel, morning and evening. The license here is not an instrument of choice, it is an instrument of obligation.

The remaining lengthening effects fill in the household's economic position. Family income up to 100,000 BDT adds 1.12 kilometers, and the direction of this effect deserves emphasis. It is the moderate-income joint families, not the affluent ones, that sit far from the activity concentrations, because the affluent joint household occupies inherited central property while the moderate one has been priced outward, the same affordability sorting seen in the child-rearing segment now operating on a larger household. Each additional minor adds 0.42 kilometers, the grandchildren's schools tugging the collective residence away from the respondent's own PAL. Rent in the 10,000–50,000 BDT band adds 0.41 kilometers, the mid-rent echo. Residence in a multi-storied building adds 0.26 kilometers, and marriage adds 0.34 kilometers and marriage adds 0.34 kilometers, reversing its sign yet again relative to the child-rearing segment. The reversal is informative rather than puzzling. The married member of a joint household is precisely the mid-generation earner described above, whose own proximity is subordinated to the extended household's location, whereas in the nuclear family the married earner is the center around which the household organizes.

The compressing forces are few, and each is a small act of relief within the collective constraint. Self-employment shortens the commute by 0.67 kilometers, the home-proximate enterprise once again offering the escape that salaried work cannot. Each additional household member shortens it by 0.10 kilometers, suggesting that as the household grows it gravitates marginally toward locations that minimize the sum of many journeys rather than any single one.

Standing back from the three figures, a coherent behavioral hierarchy emerges and it can be stated in a sentence. Commute distance in Dhaka is lengthened by motorization and by the pursuit of affordable space and it is shortened by marriage-centered centralization, multi-earner coordination, self-employment and settled tenure with the strength, and sometimes the very sign, of each force set by the structure of the household in which it operates. It is this conditional character of the distance mechanisms, invisible to any model that pools households together, that constitutes the central empirical payoff of estimating the distance equation jointly with the living arrangement choice.

4.5 Policy Recommendations

The empirical results carry direct implications for housing, transportation, and land-use policy in Dhaka. The recommendations below are organized around the five strongest

behavioral mechanisms identified by the model: the affordability–commute trade-off, the transit orientation of large households, the differentiated housing demand of distinct household structures, the accessibility channel operating through spatial context and transit orientation, and the distance-enabling effect of motorization.

4.5.1 Affordable Rental Housing near Primary Activity Concentrations

The single most consistent economic finding of the study is that households seeking housing in the dominant mid-rent band (10,000–50,000 BDT per month) commute significantly farther, with the effect exceeding one kilometer for families with children. The policy counterpart is the expansion of affordable rental supply in proximity to the major employment and educational concentrations that constitute most respondents' PALs. Instruments available to RAJUK and the city corporations include density bonusing and floor-area-ratio relaxation conditioned on affordable rental units, inclusionary requirements in redevelopment around employment centers, regularization and upgrading of informal rental supply and targeted public–private rental construction for the middle-income segment whose affordability stress is well documented in the Dhaka literature (Giti, 2018; Jahan Heme et al., 2020; Kamruzzaman & Ogura, 2008). Because 73 percent of sampled households rent, interventions on the rental margin will reach far more households than ownership-oriented programs. International evidence confirms that locating affordable housing in accessible locations simultaneously reduces household transportation burdens and vehicle travel (Howell et al., 2018; Makarewicz et al., 2020), and the residual-income affordability logic implies that the policy metric should combine housing and transport costs rather than rent alone (Dewita et al., 2020).

4.5.2 Transit-Oriented Development around MRT Stations

The model identifies joint families, Dhaka's largest and most vehicle-poor household structures, as the constituency most strongly oriented toward MRT, but only under walkable station access: willingness to use MRT within 500 meters raises joint family probability by 24 percentage points, while access beyond 1.2 kilometers reduces it by 28 points. Moreover, for households stating willingness to use the MRT, better modal accessibility uniformly and significantly compresses commute distance across all family segments. These results argue for transit-oriented development of the station areas along Line 6 and the lines under construction, with three specific emphases. First, land use within the 500-meter walk shed should prioritize larger family-format dwellings, not only compact units, because the marginal transit-oriented household is multigenerational.

Second, the quality of the pedestrian first mile, including continuous footpaths, safe crossings, lighting, and weather protection, determines whether the 500-meter behavioral threshold is achievable in practice, a point underscored by RUM-based transit accessibility research showing that first-mile conditions dominate perceived access (Hasnine et al., 2019). Third, feeder integration is needed to extend the effective catchment beyond 1.2 kilometers, where the model shows household transit orientation collapsing; organized rickshaw and paratransit feeder arrangements with integrated fares are the realistic near-term instrument in Dhaka's street environment.

4.5.3 Accessibility-Targeted Density and Mixed Use

Households in dense and residential neighborhoods commute shorter distances when transport supply is adequate, indicating that density and accessibility are complements rather than substitutes. Planning policy should therefore treat density and residential amenity as complements of, rather than substitutes for, accessibility: upzoning around activity concentrations should be paired with protected residential environments, neighborhood services, and open space, the multi-dimensional residential desirability that over 86 percent of respondents rated as important in the attitudinal data of CHAPTER 3. The Detailed Area Plan's land-use framework provides the regulatory vehicle; the model results suggest its implementation should concentrate mixed-use intensification where modal accessibility is demonstrably high, in line with integrated land-use–transport planning experience elsewhere in the Global South (Guan & Peiser, 2018; van Heerden et al., 2022; Waldeck et al., 2020).

4.5.4 Housing Supply Responsive to Household Structure

The living arrangement results document sharply differentiated dwelling demand across household structures. Non-family households concentrate in units below 1,300 square feet, nuclear families in intermediate sizes within the mid-rent band, and joint families in larger units, with the presence of multiple seniors raising joint family probability by 36 percentage points. Housing policy and building regulation should accommodate this full distribution rather than converging on a single unit format. Three implications follow. First, the mess and shared-accommodation sector that houses the non-family workforce deserves regularization and minimum-standard regulation rather than suppression, since it is the entry housing of Dhaka's educated migrant labor force. Second, family-format rental units in the mid-rent band near schools would directly relieve the child-anchored commute lengthening identified in Section 4.4.2; coordinating residential intensification

with the distribution of quality schools, or strengthening transport links to existing schools, addresses the same mechanism from the education side (Ewing et al., 2004). Third, large multigenerational units, including designs with semi-independent elder quarters, should be enabled in transit-rich locations, simultaneously serving the elderly-care function that joint families perform and the transit ridership they supply.

4.5.5 Managing Motorization and Commute Lengthening

Across two of the three family segments, a driving license lengthens commutes by 1.2 to 1.7 kilometers, the largest individual-level distance effects in the model. As incomes rise and motorization spreads, this mechanism will lengthen commutes and feed peripheral expansion unless accessible alternatives keep pace, the canonical sprawl dynamic of motorizing cities (Brueckner & Fansler, 1983; Yao & Wang, 2014). The policy response is not license restriction but the provision of competitive alternatives at the locations motorized households would otherwise abandon: extension of trunk transit, organized feeder paratransit, and, given the estimated VTTS of roughly 200 BDT per hour, travel-time-competitive service design rather than fare measures alone. The high alternative-specific constants of door-to-door paratransit modes further indicate that comfort, reliability, and crowding, rather than fares, are the qualities on which public transport currently loses utility. Investment in these service dimensions will yield the largest accessibility gains per the structure of the estimated mode choice component. Finally, the consistent negative accessibility interactions imply that every improvement in modal accessibility delivers a double dividend, both shifting mode choice and compressing the residential footprint of accessibility-motivated households.

4.6 Summary

This chapter presented and discussed the estimation results of the joint mode choice–living arrangement–commute distance model for Dhaka. The model attains a strong statistical fit (adjusted rho-squared of 0.83 against the null and 0.69 against the constants-only benchmark with 111 parameters), and its mode choice component yields behaviorally credible level-of-service sensitivities implying a value of travel time savings of approximately 200 BDT per hour. The copula correlation between the unobserved components of living arrangement and commute distance is positive (0.29), revealing that in Dhaka the latent forces of family-centered household formation pull households toward longer, not shorter, commutes.

The living arrangement results portray household structure as the joint outcome of life cycle, culture, and housing economics: age and singlehood embed adults in family and joint family households. Multiple earners enable nuclear independence whereas multiple seniors drive multigenerational co-residence. Floor area, tenure, and rent sort the arrangements across distinct housing submarkets. The marginal effects quantify these forces, with singlehood shifting up to 47 percentage points of probability away from child-rearing nuclear families and multiple seniors shifting 36 points toward joint families. The distance results identify motorization and the pursuit of affordable mid-rent housing as the principal commute-lengthening forces, with effects of 1.1 to 1.7 kilometers, and marriage-centered centralization, multi-earner coordination, and self-employment as the principal compressing forces. The accessibility interactions establish that, for households situated in dense or predominantly residential environments and, above all, for those willing to use the MRT, better modal accessibility significantly shortens commutes across all family segments. The policy discussion translated these mechanisms into five directions: affordable rental supply near activity concentrations, transit-oriented development with walkable first miles and family-format housing, accessibility-targeted densification, household-structure-responsive housing supply, and the management of motorization through competitive alternatives. The next chapter draws the overall conclusions of the thesis, states its contributions, and outlines directions for future research.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Research Summary

This thesis investigated how urban residents of Dhaka jointly decide with whom to live and how far to live from their Primary Activity Location, and how the perceived performance of the transport system mediates the trade-off between these two long-term decisions. Motivated by the near absence of living arrangement modeling in developing megacities and by the affordability and accessibility pressures characteristic of Dhaka's housing and transport markets, the study developed a joint econometric model in which a twelve-alternative multinomial logit mode choice component supplies an expected maximum utility (logsum) measure of modal accessibility to a four-alternative living arrangement choice model and a segment-specific lognormal commute distance model, with the latter two components linked through a Gaussian copula capturing their correlated unobserved factors. The model was estimated by full-information maximum likelihood on 927 valid observations from the purpose-designed household survey, enriched with Google Routes API travel skims and RAJUK Detailed Area Plan land-use data, and achieved an adjusted rho-squared of 0.83.

The empirical results establish, first, that living arrangement in Dhaka is the joint product of life cycle, cultural institutions, and housing economics. Age and singlehood embed adults within family and joint family households, the reverse of the pattern documented in Western contexts where singlehood and independent living coincide; multiple earners are the enabling condition of independent nuclear family formation; and the presence of multiple seniors is the single strongest determinant in the model, raising the probability of joint family living by 36 percentage points and confirming multigenerational co-residence as the city's de facto elderly-care institution. Dwelling size, tenure, and rent sort the four arrangements into distinct housing submarkets, from the sub 1,300-square-foot units of the non-family sector through the mid-rent rental mainstream of nuclear families to the large dwellings of joint households.

Second, the results establish that commute distance in Dhaka is structured by household arrangement and lengthened, above all, by motorization and the pursuit of affordable housing. A driving license lengthens commutes by 1.2 kilometers among nuclear families

without children and 1.7 kilometers among joint families, while occupying housing in the dominant 10,000–50,000 BDT rent band lengthens the commute of families with children by 1.1 kilometers. Households purchase attainable rents and dwelling space with travel. The compressing forces are equally systematic: marriage-centered centralization in child-rearing households, multi-earner coordination, long residential tenure and self-employment constraints all shorten commutes. The positive copula correlation indicates that beyond these observed mechanisms, the latent inclination toward family-oriented living itself carries a willingness to commute farther, a finding opposite in sign to the western household culture (Hasnine et al., 2023) and diagnostic of a city in which family-centered housing consumption, rather than proximity, anchors residential choice.

Third, and central to the thesis, the results establish that modal accessibility shapes these residential outcomes in a behaviorally differentiated way. Accessibility enters the living arrangement margin only weakly and indirectly, but on the distance margin it operates powerfully through residential motivation: for households residing in high-density areas or residential land-use zones, and for those stating willingness to use the MRT, greater expected maximum utility of mode choice significantly compresses commute distance, with the MRT-willingness channel significant and nearly uniform across all three family segments. The stated MRT willingness results sharpen this picture into a threshold: joint families, the largest and least motorized household structures, are strongly transit-oriented within a 500-meter walk of stations and decisively not so beyond 1.2 kilometers. In Dhaka, rail transit proximity plays the role that walking and cycling accessibility play in North American settings: it is the accessibility channel through which the transport system can pull households closer to their activity locations.

Taken together, these findings answer the research questions posed in CHAPTER 1. Living arrangement and commute distance in Dhaka are not separable decisions: they are correlated through both observed accessibility and latent household orientation, and any housing, transport, or land-use policy that treats them separately will misestimate its own consequences. The estimated value of travel time savings of approximately 200 BDT per hour, the affordability-driven commute penalties, and the household-structure-specific transit thresholds together provide the quantitative basis on which such integrated policy can be designed.

5.2 Research Contribution

The contributions of this thesis are methodological, empirical, and practical.

5.2.1 Methodological Contribution

The thesis develops, to the author's knowledge, the first joint discrete–continuous model of living arrangement and commute distance for a developing-country megacity, extending the copula-linked framework of (Hasnine et al., 2023) from a student population in a high-income, transit-mature city to the general adult population of Dhaka. The extension is not merely geographic. The model generalizes the living arrangement space from student configurations to the culturally specific structures of South Asian households, including the joint family and expands the mode choice universe from five to twelve alternatives spanning Dhaka's full paratransit spectrum. The study defines the continuous outcome relative to a Primary Activity Location encompassing both work and education anchors and implements the expected maximum utility exclusively through interactions with non-mutually-exclusive indicators of objective spatial context, stated transit willingness, and household mobility endowments, allowing the accessibility sensitivity of distance to vary with the measured context in which accessibility is experienced. The closed-form copula likelihood, programmed in GAUSS and estimated by full-information maximum likelihood, demonstrates that this rich joint structure is estimable on a modest survey sample, providing a transferable template for similar analyses in other data-constrained developing-city contexts (Hawkins & Habib, 2021; Zhang et al., 2019).

5.2.2 Empirical Contribution

Empirically, the thesis supplies the first behavioral estimates of the accessibility that is living arrangement and distance relationship for Dhaka, a city whose residential location literature has so far examined location choice and commute distance without the living arrangement dimension (Choudhury & Ayaz, 2015) and whose affordability literature has been largely descriptive (Giti, 2018; Jahan & Kalam, 2012). The study quantifies, for the first time in this context, the commute-distance price of affordable housing (about 1.1 kilometers for mid-rent families with children), the distance-enabling effect of motorization (1.2–1.7 kilometers per license), the elderly-care basis of joint family formation (36 percentage points), the reversal of the singlehood–independence relationship relative to Western evidence, the walkable-access threshold of joint family transit orientation, and a positive rather than negative latent correlation between family-

oriented living and commute length. It also contributes one of the first revealed-preference estimates of the value of travel time savings for general commuting in Dhaka, at approximately 200 BDT per hour. Beyond the model, the thesis contributes the main survey itself, a validated, geocoded dataset of 927 households across 53 thanas combining socio-demographics, household composition, housing attributes, attitudinal motivations, and route-level travel skims, which constitutes a reusable empirical foundation for residential and travel behavior research in Bangladesh.

5.2.3 Practical and Policy Contribution

For practice, the thesis converts its behavioral estimates into an evidence base for integrated housing–transport–land-use planning in Dhaka at the moment the MRT network is being built out. The household-structure-specific MRT access thresholds identify both the constituency (large, multigenerational, vehicle-poor households) and the geometry (a genuinely walkable 500-meter shed, feeder integration beyond 1.2 kilometers) of transit-oriented development. The affordability–commute estimates justify locating mid-rent family housing near activity concentrations and support the use of combined housing-plus-transportation affordability metrics in plan evaluation (Dewita et al., 2020; Makarewicz et al., 2020; Stone & Stone, 2010) and the estimated VTTS and mode-specific constants indicate that travel time, reliability, and comfort, rather than fares alone, are the margins on which public transport competes for Dhaka's commuters. These quantities are directly usable in the appraisal frameworks of RAJUK's Detailed Area Plan implementation and of the city's transit investment programs (van Heerden et al., 2022; Waldeck et al., 2020).

5.3 Direction for Future Research

Several extensions follow naturally from the limitations of the present study. First, the analysis rests on cross-sectional data, so the estimated relationships describe equilibrium associations rather than dynamic adjustments. The sample of 927 observations, while sufficient for stable estimation, is also modest, so collecting a larger dataset would permit finer living arrangement categories and more disaggregated analysis. Longitudinal residential-history data would, in addition, allow the modeling of living arrangement transitions and relocation timing over the life course, distinguishing the milestone-triggered moves emphasized in the mobility biography literature from the steady-state sorting captured here (Mulder & Clark, 2002; Rashidi & Mohammadian, 2015).

Second, the continuous distance formulation, while parsimonious and well suited to the available data, abstracts from the spatial identity of alternative residential locations. A discrete location choice extension over finer zones, with sampling of alternatives and zone-level housing-market attributes, would recover the spatial distribution of demand and permit the simulation of zone-specific housing and transit interventions. Latent-auction and place-of-residence–place-of-work formulations offer promising routes under Dhaka's data constraints (Hawkins et al., 2021; Zhang et al., 2019).

Third, the dependence structure could be enriched. The present model employs a single Gaussian copula between living arrangement and distance. Alternative copula families with tail dependence, segment-specific dependence parameters, or a full trivariate structure incorporating the mode choice margin would test the robustness of the positive latent correlation identified here. Relatedly, allowing random taste heterogeneity in the level-of-service and accessibility parameters, for example through mixed logit or latent-class formulations, would yield distributional rather than average values of travel time savings and accessibility sensitivity, of direct use in equity-aware appraisal.

Fourth, the attitudinal content of the conducted survey invites an integrated choice and latent variable (ICLV) extension in which transit orientation and residential preferences are treated as latent constructs underlying the stated MRT-willingness responses and attitudinal importance ratings, addressing potential endogeneity between stated orientations and realized locations. This connects the model to residential self-selection literature (Schwanen & Mokhtarian, 2005). As the MRT network builds out, the stated willingness indicator can also be benchmarked against revealed station-area ridership.

Fifth, the policy analyses sketched in chapter 4 can be formalized. The estimated model supports logsum-based welfare evaluation of accessibility scenarios, disaggregated values of travel time savings by income group, willingness-to-pay measures for walkable MRT access, and the construction of a housing-plus-transportation affordability index for Dhaka's thanas, implementing these on the estimated parameters, and validating them as the MRT network expands, is a natural next step. Finally, replication in other Bangladeshi and South Asian cities, and with larger samples permitting finer living arrangement categories such as the separate identification of mess households and single-person households, would establish the external validity of the behavioral regularities documented in this thesis and consolidate a Global South evidence base for the joint modeling of who people live with, where they live, and how they travel.

Finally, and most importantly for the external validity of the household typology, the non-family category merits dedicated study. Sample size obliged the present analysis to merge two behaviorally distinct groups, people living entirely alone and people sharing messes, sublets, or hostel rooms, into a single reference alternative. These are not equivalent in Dhaka, and the difference is culturally specific. In much Western research, living alone is the normative expression of early-adult independence, while shared non-family living is transitional or economic (Mulder & Clark, 2002). In Dhaka the relationship is close to inverted: living truly alone is uncommon, often discouraged, and costly, whereas shared mess and hostel accommodation is the dominant, and frequently the only feasible, route by which unmarried students and migrant workers secure accessible housing, in a market where landlords openly prefer families over so-called bachelors. A larger, purpose-designed survey retaining enough single-person and shared-household observations to estimate these as separate alternatives would test whether the solo dweller and the mess sharer respond differently to rent, accessibility, and transit proximity, sharpening the contrast with the Western literature on leaving home and strengthening the Global South evidence.

In closing, this thesis has shown that in Dhaka the decisions of whom to live with, where to live, and how to travel are jointly determined, with modal accessibility central to all three. By quantifying these linkages within a transferable joint model, the study provides both a behavioral foundation and a practical evidence base for integrated housing, land-use, and transport planning as the city's mass-transit network takes shape.

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APPENDIX

A.1 Pseudocode Framework (for the Joint model estimation in GAUSS)

BEGIN JOINT MODEL ESTIMATION

LOAD cleaned analytical dataset

RECODE living arrangement variable into four categories

category 1 = non family household

category 2 = family without children

category 3 = family with children

category 4 = joint family

DEFINE number of observations

DEFINE constant vector

DEFINE joint parameter vector

FUNCTION log likelihood(parameter vector, dataset)

INITIALIZE likelihood components

PART A: MODE CHOICE MODEL

DEFINE twelve travel mode alternatives

FOR each respondent

FOR each available travel mode

COMPUTE mode utility as a function of

alternative specific constant

travel time

travel cost

END FOR

COMPUTE denominator of multinomial logit model

COMPUTE probability of each travel mode
COMPUTE expected maximum utility from available modes
END FOR

PART B: LIVING ARRANGEMENT MODEL

DEFINE four living arrangement alternatives
SET non family household as the reference category
FOR each respondent
 COMPUTE utility of family without children using
 individual characteristics
 household characteristics
 housing attributes
 mobility resources
 built environment variables
 modal accessibility
 COMPUTE utility of family with children using
 individual characteristics
 household characteristics
 housing attributes
 mobility resources
 built environment variables
 modal accessibility
 COMPUTE utility of joint family using
 individual characteristics
 household characteristics
 housing attributes

mobility resources

built environment variables

modal accessibility

COMPUTE living arrangement choice probabilities

END FOR

PART C: COMMUTE DISTANCE MODEL

DEFINE commute distance as distance from residence to Primary Activity
Location

APPLY lognormal distance specification

FOR each respondent

IF living arrangement is family without children THEN

COMPUTE expected log distance using

individual characteristics

household characteristics

housing attributes

mobility resources

built environment variables

modal accessibility interactions

ELSE IF living arrangement is family with children THEN

COMPUTE expected log distance using

individual characteristics

household characteristics

housing attributes

mobility resources

built environment variables

```

        modal accessibility interactions
ELSE IF living arrangement is joint family THEN
    COMPUTE expected log distance using
        individual characteristics
        household characteristics
        housing attributes
        mobility resources
        built environment variables
        modal accessibility interactions
ELSE
    RETAIN non family household as reference segment
END IF
END FOR

```

PART D: COPULA BASED JOINT STRUCTURE

```

    DEFINE common distance scale parameter
    DEFINE copula correlation parameter
    FOR each respondent
        TRANSFORM living arrangement probability into normal scale
        STANDARDIZE observed log distance
        COMPUTE lognormal distance density
        COMBINE living arrangement probability and distance density through
Gaussian copula
    END FOR

```

PART E: LIKELIHOOD CONTRIBUTION

FOR each respondent

 SELECT probability of observed living arrangement

 SELECT probability of observed travel mode

 SELECT distance density for relevant household segment

 COMPUTE individual likelihood contribution

END FOR

 COMPUTE total log likelihood as sum of individual log likelihoods

RETURN total log likelihood

END FUNCTION

PART F: PARAMETER ESTIMATION

SET starting values from standalone model estimates

INITIALIZE maximum likelihood estimation

ESTIMATE all parameters jointly

CHECK convergence criteria

COMPUTE standard errors

COMPUTE t statistics

REPORT final estimates and goodness of fit measures